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HERO

HERO FOUNDATION

The Theory of Accountable Value Creation

Yin Bo

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AUTHOR Yin Bo	CONTRIBUTORS To be confirmed
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Foundation Boundary

Hero Canon 002 is the theoretical Foundation from which HeroOS can be derived. It defines propositions, relationships, boundaries, counterclaims, and tests for accountable Value creation.

It is not the formal HeroOS specification. Canon 006 will specify how HeroOS operates, including its roles, permissions, records, states, workflows, interfaces, controls, and conformance requirements.

Intended Readers

Hero Foundation is written for founders, CEOs, product and engineering leaders, architects, domain experts, and others responsible for designing value-creation systems in the Age of Abundant Intelligence.

Foundational Thesis

Hero begins from a shift in scarcity: when intelligence and execution capability become easier to access and compose, accountable human judgment becomes more important rather than less.

The theory must show why an identifiable human center remains necessary, how composed capability can amplify that human, how value becomes testable through evidence and consequence, and how execution can become organizational learning.

How to Read This Foundation

Every major chapter has five proof obligations:

1. Definition: What exactly is being claimed?
2. Derivation: Why does the claim follow from prior premises?
3. Counterexample: What case appears to contradict it?
4. Boundary: Where does the claim stop applying?
5. Falsifiability: What evidence would require the claim to be revised or rejected?

Source IDs such as `[S1]` refer to the Research Sources appendix in this edition and the accompanying source register in the Canon source package.

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1. The Foundational Question

KEY IDEA

When capability becomes abundant, the foundational organizational question shifts from who can execute to who is accountable for value.

Purpose

Establish the question that Canon 002 must answer and distinguish it from a technology forecast, management trend, or moral slogan.

Theory

Organizations have historically treated execution capacity as a primary organizing constraint. Specialized knowledge lived in people, software was expensive to build and change, coordination required repeated human handoffs, and access to expertise depended on hiring, hierarchy, or procurement. Under those conditions, organizational design reasonably concentrated on acquiring labor, allocating tasks, supervising work, and increasing throughput.

Hero begins when that assumption stops being sufficient.

The cost and accessibility of some forms of intelligence-enabled execution are changing quickly. A person can now draw upon models, software services, platforms, workflows, and remote specialists that perform parts of research, analysis, writing, design, coding, monitoring, and coordination. This does not make intelligence infinite, equally distributed, reliable, or free. It changes the range of actions that can be attempted without first expanding a conventional team.

When the number of executable options expands, execution alone no longer determines whether an organization creates Value. The organization must still decide which problem matters, whose interests count, what risks are acceptable, which Capability can be trusted, what Evidence would change the judgment, and who must answer for the result. More possible action can increase the cost of poor selection. It can also allow weak assumptions, hidden bias, and unowned consequences to scale faster.

The foundational unit of Hero is therefore not the task, the prompt, the agent, the team, or the output. It is the `Value Commitment`: an accountable promise by an identifiable human to pursue defined Value within explicit boundaries and to evaluate the resulting Outcome and Consequences.

This changes the location of human importance. Human value does not depend on personally executing every step. It depends on purpose, legitimate interpretation, responsible judgment, the governance of composed Capability, and answerability for material Consequences. A Hero may be surrounded by extensive non-human and human execution, but the Value Commitment must not become ownerless.

The human center is necessary not because human judgment is always correct. It is necessary because Value is contestable, trade-offs affect different people, and responsibility must remain attributable when a system acts under uncertainty. NIST's AI Risk Management Framework similarly treats governance as cross-cutting and emphasizes human centricity, social responsibility, sustainability, Context, and attention to intended and unexpected impacts across AI actors.[S5] Hero goes further by proposing that each Value Commitment retain a named Primary Accountability center.

Derivation

The foundational argument can be stated as a sequence:

1. Intelligence-enabled execution becomes more accessible and composable in a growing set of domains.
2. Greater execution capacity increases the number and speed of possible actions.
3. Possible action does not determine legitimate Value.
4. Value requires interpretation, trade-offs, Evidence, and acceptance of material Consequences.
5. These judgments require an identifiable Accountability center.
6. Therefore, increasing execution abundance raises rather than removes the need for accountable human judgment.

This is a theory of organizational constraint, not a claim of human perfection. The Hero is answerable for responsible judgment within an accepted scope, not guaranteed success or omnipotence.

Claims to Establish

- Hero is a response to a shift in the economics and accessibility of intelligence.
- The relevant unit of analysis is accountable value creation, not task completion alone.
- Human importance moves from exclusive execution toward purpose, judgment, accountability, and consequence.

Required Proof

- Define abundant intelligence as increased accessibility and composability, not infinite or free intelligence.
- Show why more execution capacity does not automatically create more value.
- Explain why value creation needs an accountable center under ambiguity and competing interests.

Counterexample to Address

A fully automated system may appear to produce useful outcomes without an identifiable human making each decision.

Boundary

The theory does not claim that every individual execution requires synchronous human intervention.

Falsification Test

The claim weakens if durable, legitimate accountability for material consequences can be borne by a non-human system without attribution to any human or human institution.

2. The Age of Abundant Intelligence

KEY IDEA

Abundance means capability is increasingly accessible, reusable, and composable, not that scarcity has disappeared.

Purpose

Define the environmental premise of Hero precisely enough to be challenged.

Theory

Abundance in Hero is a relative and operational concept. It means that a useful class of intelligence or execution becomes easier to access, less expensive to invoke, faster to reuse, and more practical to combine. It does not mean that all capability is available, that quality is assured, or that marginal cost has reached zero.

The distinction matters because four different things are often collapsed into the word "intelligence":

- **Intelligence** is the capacity to interpret, infer, generate, or decide within some domain.
- **Capability** is a repeatable ability to produce a verifiable result under stated conditions.
- **Execution Capacity** is the available quantity, speed, and concurrency with which qualified Capability can be exercised.
- **Outcome** is the change that results in the world and must be evaluated against Value.

Lower inference cost is Evidence of changing access economics, not proof of Value. Stanford's 2025 AI Index reports that the cost of querying a model at a GPT-3.5-equivalent MMLU performance level fell from USD 20 per million tokens in November 2022 to USD 0.07 by October 2024, a reduction of more than 280 times.[S1] That change makes some machine intelligence easier to borrow. It does not show that the borrowed capability is qualified for a particular Context, that the organization can integrate it, or that the resulting Outcome is beneficial.

Real-world productivity evidence is heterogeneous. An NBER study of 5,179 customer-support agents found a 14% average increase in issues resolved per hour after access to a generative AI assistant, including a 34% improvement for novice and lower-skilled workers but minimal impact on experienced and highly skilled workers.[S2] The same tool category therefore changed execution capacity differently depending on worker experience and task Context.

Counterevidence is equally important. METR's randomized early-2025 study found that experienced open-source developers working in repositories they knew well took 19% longer when AI tools were permitted.[S3] METR explicitly warned against generalizing that result to most software development. Its 2026 follow-up then reported that wider AI adoption and selection effects made later estimates unreliable; the researchers considered greater speedup likely but described the magnitude as weakly evidenced.[S4]

Together, these findings support neither universal abundance nor universal failure. They support a Context-sensitive transition: access cost and available machine capability can improve rapidly while realized execution benefits remain dependent on experience, task structure, implicit requirements, quality thresholds, integration practice, and time.

Indicators of Abundance

A domain is moving toward execution abundance when several indicators improve together:

- the cost of accessing a defined Capability falls;
- the latency between intent and qualified execution falls;
- Capability can be reused across commitments with limited reinvention;
- multiple Capability Units can be composed without disproportionate coordination cost;
- qualification and Evidence can keep pace with increased execution;
- the organization can add execution capacity without equivalent growth in headcount.

No single indicator is sufficient. Cheap unqualified output is not abundant Capability. Fast execution without observability is not governable scale. High benchmark performance without Context fit is not reliable organizational capacity.

Uneven Abundance

The Age of Abundant Intelligence arrives unevenly. Digital knowledge work may experience falling access cost while physical work remains constrained by equipment, energy, materials, location, safety, or human presence. A regulated decision may be technically executable but legally non-delegable. A relationship-intensive decision may require trust that cannot be substituted by output quality. An organization may have access to strong models yet lack clean data, permission, integration, or governance.

Abundance is therefore a property of a defined Capability in a defined Context at a defined time. It must never be inferred from the existence of a technology alone.

Claims to Establish

- Intelligence-enabled execution is becoming cheaper and more widely accessible in many knowledge domains.
- Software, AI, platforms, services, and specialists can increasingly function as Capability Units.
- Abundance is uneven across domains, geographies, organizations, and physical constraints.

Required Proof

- Separate intelligence, capability, execution capacity, and successful outcome.
- Identify measurable indicators of abundance such as access cost, latency, reuse, and composition speed.
- Explain why physical, regulated, safety-critical, and relationship-intensive domains may retain execution scarcity.

Counterexample to Address

In physical operations, scarce equipment, materials, permissions, or skilled labor may remain the binding constraint.

Boundary

Hero does not assert that execution is never a bottleneck. It asserts that the relative scarcity pattern is changing and must be tested by domain.

Confirmed Scarcity-Shift Hypothesis

"Execution is no longer the bottleneck" remains valid as Manifesto language. Canon 002 uses the bounded theoretical claim:

> Execution is no longer the universal or default bottleneck. In a growing class of intelligence-enabled work, the binding scarcity is shifting toward Value judgment, Accountability, governance, Evidence, and Consequence.

The hypothesis applies when required Capability is accessible, composable, sufficiently qualified, and verifiable; execution cost or latency is falling; necessary data, tools, and access exist; and physical, legal, safety, or relationship constraints do not remain dominant.

Execution may remain the primary bottleneck in physical production, field service, scarce equipment or labor, safety-critical and regulated work, trust-intensive work, exploratory work without reusable Capability, and organizations lacking data, access, or infrastructure.

For a Value Commitment, the practical test is:

> ### If qualified execution capacity doubled, would the intended Outcome materially improve?

If yes, Execution may remain the binding constraint. If no, the constraint may instead be unclear Value, insufficient Authority, ambiguous Accountability, ungovernable Capability, weak Evidence, or delayed judgment.

The Scarcity-Shift Hypothesis states that as accessible and composable execution capacity expands, organizational constraint increasingly moves from whether work can be executed to what is worth executing, who accepts Accountability, and how results are judged.

Falsification Test

The broad claim must be narrowed if execution capacity remains the dominant constraint across the domains for which Hero is proposed.

3. The Scarcity Shift: Execution to Accountability

KEY IDEA

As execution capacity expands, the scarce capacity becomes answerable judgment about value and consequence.

Purpose

Derive the statement “Accountability is the new scarcity” rather than use it only as a declaration.

Theory

Scarcity shifts when one constraint expands faster than another. If qualified execution becomes easier to obtain while the capacity for legitimate judgment, clear commitment, governance, and consequence ownership does not expand at the same rate, Accountability becomes the binding organizational constraint.

This does not mean that fewer people are responsible. It means that delegation multiplies the need to know where final answerability resides. A single Hero may use many Capability Units, and many people may hold Responsibilities or Independent Accountabilities. The problem begins when the system can act but cannot clearly answer four questions:

1. What Value was the action intended to create?
2. Who had Authority to make the material judgment?
3. What Evidence could confirm or challenge the decision?
4. Who must explain and respond to the Consequences?

Technology does not resolve these questions by producing more output. In fact, greater execution capacity can make them harder because more actions can be launched, combined, and repeated before their effects are understood. NIST's AI RMF describes governance as a cross-cutting function and notes that different actors across an AI lifecycle may hold different risk perspectives and responsibilities.[S5] Hero accepts this plurality but rejects the conclusion that plurality should make the Value Commitment ownerless.

Responsibility and Accountability

Responsibility and Accountability are related but not interchangeable.

Responsibility concerns the obligation to act, perform, support, review, or ensure that work occurs. It can be divided because complex work requires many contributors and controls.

Accountability concerns final answerability within a defined scope for judgment, Value, and material Consequences. It remains identifiable because delegation cannot answer the question of why a Value Commitment was accepted, how conflicting Evidence was interpreted, or why a risk was accepted.

The non-transferability claim does not mean that one person absorbs all legal, moral, professional, or operational responsibility. It means that execution cannot be used to dissolve the Primary Accountability attached to a Value Commitment. Independent Accountability remains necessary where safety, compliance, professional integrity, or constitutional limits require separate judgment.

Why Accountability Becomes Scarce

Execution can often be replicated through capital, software, platforms, external services, or additional Capability Units. Accountable judgment is harder to scale because it requires Context, legitimate Authority, attention, interpretation, and willingness to accept material Consequences.

The scarcity is therefore not the number of people willing to receive a title. It is the capacity to make an explicit commitment, maintain enough Context to judge it, resist misleading output, respond to counterevidence, and remain answerable when outcomes are uncertain or adverse.

A Hero organization does not solve this scarcity by naming everyone accountable. Undifferentiated shared Accountability makes the location of judgment less visible. The system must instead distinguish Primary Accountability, Distributed Responsibility, Independent Accountability, and Institutional Accountability.

Collective Structures

Boards, committees, partnerships, and legal entities can carry real institutional authority and responsibility. Hero does not deny collective governance. It requires collective structures to preserve identifiable decision rights and human role-holders.

A committee may deliberate collectively, but its mandate must state how a decision becomes authorized, which Value Commitment it governs, how dissent and conflicts are handled, and who must ensure that Evidence and Consequences receive a response. An institution may remain legally accountable while a named Hero holds Primary Accountability for a specific Value Commitment. These layers can coexist if their scopes are explicit.

The alternative is not collective governance versus individual heroism. The alternative is accountable collective governance versus structures in which answerability disappears between roles.

Claims to Establish

- Delegating execution does not delegate final answerability.
- More possible actions increase the need for selection, interpretation, and consequence ownership.

- Accountability cannot be inferred from title, authority, or proximity to execution.

Required Proof

- Test the confirmed distinction between Responsibility and Accountability against real organizational structures.
- Define what makes accountability identifiable and non-transferable within a stated scope.
- Address individual, shared, institutional, legal, and moral forms of accountability.

Confirmed Shaping Baseline

Responsibility is the obligation to act, perform, or ensure that defined work occurs. Responsibility may be allocated, delegated, or shared.

Accountability is final answerability, within a defined scope, for judgment, value, and material consequences. Execution may be delegated, but Accountability must not disappear through delegation.

Each Value Commitment requires one identifiable **Primary Accountability** center: a named human Hero. This does not make all other accountability disappear.

- **Distributed Responsibility** may be assigned across people, teams, and Capability Units.
- **Independent Accountability** may be held by other humans for distinct safety, risk, compliance, professional, or constitutional constraints.
- **Institutional Accountability** may be borne by an organization in law or governance, but must be realized through explicit roles and identifiable human role-holders.
- **Shared Accountability** is valid only when scope, decision rights, and conflict resolution are explicit. Undifferentiated shared accountability, in which everyone is responsible but no one must finally answer, is not valid under Hero.

Counterexample to Address

Boards, partnerships, committees, and legal entities often carry collective or institutional accountability.

Boundary

One Primary Accountability center does not imply unilateral authority, unlimited liability, omniscience, or the removal of independent governance. It means the Value Commitment cannot lose its identifiable human answerability inside a collective structure.

Accountability is bounded by the accepted Value Commitment, legitimate decision authority, reasonable foreseeability, practical ability to control or influence, and the stated evaluation horizon. These boundaries do not permit a Hero to accept a commitment while knowingly lacking the authority required to govern it.

Falsification Test

The non-transferability claim must be revised if accountability can be divided so completely that no identifiable human remains answerable for judgment and material consequence.

4. The Human Center of Value

KEY IDEA

A Value Commitment requires an identifiable human center because value is interpreted before it is produced.

Purpose

Explain human primacy without claiming that humans must perform every task or are always better executors.

Theory

Value begins neither in production nor in measurement. It begins when someone distinguishes an improvement worth pursuing from a change that is merely possible. Output describes what was produced. Efficiency describes a relationship between inputs and outputs. Revenue records an exchange. Preference records what an actor says or reveals that they want. Each may provide Evidence of Value, but none is sufficient by itself. A system can efficiently produce an unwanted output, earn revenue by transferring harm to another party, or satisfy an immediate preference while damaging the beneficiary's longer-term interests.

For this reason, Value is a normative and empirical judgment at the same time. It is normative because a purpose, beneficiary, and acceptable distribution of consequence must be chosen. It is empirical because the claimed improvement and its consequences must remain open to Evidence. Hero therefore rejects two reductions: that Value is whatever a decision-maker personally prefers, and that Value is whatever a metric can optimize. Personal preference lacks legitimate authority over all affected parties; an optimization metric inherits the omissions, time horizon, and stakeholder assumptions of the objective given to it.

The human center of Value is the location at which these assumptions become answerable. It need not be the person performing the work, calculating the optimum, or approving every local decision. It is the human Primary Accountability center that can explain why the beneficiary is legitimate, why this improvement matters, which competing interests were considered, what consequences were accepted, and what would cause the Commitment to be revised or stopped. An institution may authorize and constrain this judgment through law, mission, fiduciary duty, professional standards, democratic mandate, or governance. Those institutions extend the legitimacy of human judgment; they do not remove the need for an identifiable human center within a particular Value Commitment.

Beneficiaries and Legitimacy

A legitimate beneficiary may be a person, an organization, or society. These categories are not mutually exclusive and do not imply equal benefit. A product may create customer utility, employee opportunity, organizational income, supplier demand, public tax revenue, and environmental cost at the same time. The existence of several beneficiaries therefore does not settle whether the net result is legitimate. It creates a judgment problem.

Legitimacy is established through four connected questions:

1. **Purpose**: Is the improvement relevant to the authorized purpose of the Value Creation System?
2. **Standing**: Which people, organizations, or public interests are materially affected and therefore entitled to consideration?
3. **Boundaries**: Which legal, constitutional, contractual, ethical, professional, or institutional limits constrain the choice?
4. **Consequence**: Are material benefits, burdens, risks, reversibility, and time horizons made visible rather than displaced outside the Value claim?

These questions do not yield an automatic formula. They discipline Judgment. Stakeholder interests frequently conflict, and declaring every stakeholder essential does not make the conflict disappear. The Business Roundtable's statement on corporate purpose, for example, commits signatory companies to customers, employees, suppliers, communities, and long-term shareholder value. It demonstrates that a commercial conception of Value can explicitly include multiple stakeholder classes while retaining economic value, but it does not prove that their claims are naturally aligned or provide a method for resolving every trade-off. [S6]

When interests conflict, the Hero must not conceal the conflict inside a composite score. The Hero should identify whose condition is expected to improve, whose condition may worsen, which constraints are non-negotiable, what distribution is being accepted, and which Evidence would require reconsideration. A technically superior optimum may still be illegitimate if the objective excludes parties with standing, treats a protected boundary as a tradable variable, or externalizes material consequences beyond the evaluation horizon.

Economic Sustainability

Economic sustainability is not identical to short-term profit, but neither is it optional for a commercial system. A Commitment may Create, Protect, Enable, or Consume Justifiably. The four paths prevent every local initiative from being forced into an artificial direct-revenue claim while still requiring an account of resource use. Employee safety, regulatory compliance, foundational infrastructure, capability development, and the fulfillment of legitimate social obligations may consume resources without producing immediate profit. They can nevertheless be Value when their purpose, consequences, and economic relationship are explicit.

At the level of the complete commercial Value Creation System, however, the set of Commitments must retain a credible path to profit over the relevant horizon. Without that path, the organization eventually loses the capacity to continue serving any beneficiary. In a public or social system, the analogous requirement is sustainable stewardship: resources, authority, institutional trust, and operating capacity cannot be consumed without a defensible basis for renewal or prioritization. Sustainability is therefore a boundary on Value, not the sole definition of Value.

Human primacy is most important where no stable objective function can legitimately absorb all of these questions. Once a legitimate objective, operating boundary, and escalation rule have been established, machines may calculate, search, predict, or execute better than humans. Hero does not reserve execution for people. It reserves accountable interpretation of Value and accepted Consequence for a human or human-authorized institutional order.

Claims to Establish

- Value is not identical to output, preference satisfaction, revenue, or efficiency.
- Value depends on purpose, stakeholders, trade-offs, time horizon, and accepted consequences.
- The Hero is the human location where these interpretations become answerable.

Required Proof

- Define the role of human judgment under ambiguity.
- Test how legitimate value is established when stakeholder interests conflict.
- Show how legitimacy differs from technical optimization.

Confirmed Shaping Baseline

Value is a material improvement for a legitimate beneficiary: a person, an organization, or society. The improvement must be identifiable, open to evaluation through evidence, considered together with its material consequences, and supported by a sustainable economic or resource basis.

Value legitimacy comes from the relevant purpose, affected stakeholders, and institutional or constitutional boundaries. The Hero interprets, weighs, and accepts Accountability for a Value Commitment, but does not unilaterally invent what counts as Value from personal preference.

For a commercial organization, every Value Commitment does not need to produce profit directly. It must, however, explain its relationship to economic value through at least one of four paths:

- **Create**: increase revenue, margin, or profit.
- **Protect**: reduce loss, risk, liability, or loss of operating legitimacy.
- **Enable**: create capability, assets, knowledge, or options required for future value and profit.
- **Consume Justifiably**: use economic resources for legitimate human, social, legal, or institutional obligations.

At the level of the complete commercial Value Creation System, economic sustainability and a credible path to profit remain necessary. In public or social systems, the equivalent constraint is sustainable stewardship of financial, institutional, and other scarce resources.

Counterexample to Address

An optimization system may outperform human decision-makers against a stable, accepted objective function.

Boundary

Human primacy applies to accountable interpretation and consequence, not necessarily to execution quality in every task.

Falsification Test

The claim fails if contested value can be legitimately defined and revised without any human judgment or human-authorized institution.

5. Capability Theory

KEY IDEA

A Capability is a repeatable ability to produce a verifiable result under stated conditions.

Purpose

Provide the theoretical unit that allows human, software, service, platform, workflow, and AI execution to be compared without treating them as identical.

Theory

A Capability is a repeatable ability to produce a verifiable result under stated conditions. It is an ability, not the actor that possesses it; a result class, not a single output; and a qualified reliance claim, not a description of potential. Saying that a person, model, service, or platform *can* do something is therefore weaker than saying that the Capability may responsibly be used for a Value Commitment.

Five elements make the reliance claim meaningful:

- **Result** identifies the outcome class the Capability is expected to produce, including relevant quality and consequence boundaries.
- **Repeatability** means that the result can be relied upon across a meaningful class of occasions, not that every execution follows identical steps or produces identical content.
- **Verification** provides a way to distinguish acceptable results from plausible-looking failure.
- **Operating Conditions** state the Context in which the reliance claim holds, including inputs, environment, authority, dependencies, scale, time, and material assumptions.
- **Qualification** is the Evidence-backed Judgment that a particular Capability Unit is fit to provide that Capability for an intended use.

This definition separates Capability from nearby concepts. A person or tool may possess many Capabilities. A task is an instance of work that may require several Capabilities. A role groups expected Accountabilities and Responsibilities. An output is one produced artifact or action. A **Capability Unit** is the identifiable human, team, software component, AI system, service, platform, workflow, or institutional mechanism that has been qualified to execute a Capability under specified conditions.

Qualification is relational rather than absolute. A language model may be qualified to draft low-risk alternatives when a human verifies them, but not to issue a binding legal decision. A payment service may be qualified for one currency, transaction ceiling, jurisdiction, and availability target but not another. An expert may be qualified in a familiar domain yet unqualified when recency, independence, workload, or authorization requirements are not satisfied. The same Capability can therefore have multiple qualified Capability Units, and the same Unit can be qualified differently across Contexts.

NIST's systems engineering guidance defines assurance as grounds for justified confidence based on objective Evidence that reduces uncertainty to an acceptable level. It also requires the rigor of engineering activity to be commensurate with the trust placed in system elements. [S7] Hero adopts the same direction without importing a universal engineering certification scheme: confidence in Capability must be justified by Evidence proportionate to the Value Commitment and its material Consequences.

Fitness, Degradation, and Expiry

Capability is not permanently acquired. Fitness can degrade when the operating environment changes, dependencies drift, data distributions move, interfaces change, skills decay, workload exceeds limits, authority expires, or new Evidence reveals a failure mode. Qualification can also expire by design because old Evidence no longer supports current reliance. A Capability Unit that was once fit is not presumed fit forever.

The practical implication is theoretical before it is procedural: every qualification has a Context and a time boundary. Evidence of past success raises confidence only to the extent that the current use remains relevantly similar. Material Context change weakens inheritance from past Evidence and may require restriction, revalidation, or revocation. Canon 006 may later specify records, review cycles, and controls; Canon 002 establishes why some such mechanism is necessary.

Capability confidence is also bounded by verification. Where results are cheap to generate but expensive to check, apparent Capability may grow faster than reliable Capability. Where the evaluator cannot distinguish a valid result from a persuasive error, repeated production is not meaningful repeatability. Verification may be direct, sampled, independently reviewed, tested against constraints, compared with downstream outcomes, or supported by an assurance case. The method varies; the requirement to make the reliance claim challengeable does not.

Creative and Exploratory Capability

Creative, exploratory, and emergency work does not invalidate the definition. Its repeatable unit is usually not an identical output or fixed procedure. It may be the ability to generate relevant alternatives, conduct disciplined inquiry, make sound judgments under pressure, or reach an acceptable class of outcomes while respecting constraints. Verification may concern the quality of reasoning, provenance, constraint satisfaction, expert review, learning achieved, or consequences avoided rather than advance prediction of the exact answer.

The boundary remains important. A one-time success with no defensible basis for future reliance is an achievement, not yet a qualified Capability. Conversely, uncertainty about the exact output does not make an ability non-repeatable when the process of Judgment and the class of acceptable results can be evaluated. Hero's concept of Capability is broad enough for human expertise and discovery, but strict enough to distinguish responsible reliance from hope.

Claims to Establish

- Capability is distinct from a person, tool, task, role, and one-time output.

- A Capability Unit is a qualified executor of a Capability.
- Qualification depends on conditions, evidence, limits, and continued fitness.

Required Proof

- Define result, repeatability, verification, operating conditions, and qualification.
- Explain how capability degrades, expires, or becomes invalid.
- Show why the same Capability may have multiple qualified Capability Units.

Counterexample to Address

Creative, exploratory, or emergency work may not be repeatable in a narrow procedural sense.

Boundary

Repeatability may refer to a reliable class of outcomes or judgments, not identical steps or outputs.

Falsification Test

The definition must be revised if it excludes important value-creating abilities that can be responsibly relied upon but cannot satisfy any meaningful repeatability or verification condition.

6. Capability Composition

KEY IDEA

Composed capability expands what one accountable human can achieve, but composition creates governance obligations.

Purpose

Explain how scale can increase without equating scale with headcount or ungoverned automation.

Theory

Capability Composition is the deliberate arrangement of two or more Capability Units so that their combined operation can produce a result that no unit is expected to produce alone, or can produce it with greater scale, speed, quality, resilience, or reach. Composition may include people, teams, software, AI, services, platforms, workflows, controls, and institutions. Its power comes from specialization and coordination. Its risk comes from the fact that the behavior of the whole is not fully described by the qualifications of its parts.

Four relationships must be distinguished. **Composition** is the overall arrangement of Capabilities and dependencies. **Orchestration** coordinates sequence, information, decisions, and hand-offs within that arrangement. **Delegation** authorizes another actor or Unit to perform bounded work or make bounded decisions. **Governance** establishes the basis for selecting, observing, constraining, challenging, changing, and stopping the arrangement. Orchestration can be automated; delegation can be broad or narrow; governance remains necessary wherever the composition can materially affect a Value Commitment.

A composition creates new interfaces. Each interface may transform information, omit Context, delay signals, reinterpret authority, or create ambiguity about completion. It also creates dependencies: a qualified Unit may rely on an identity service, dataset, model provider, operator, network, approval, or supplier that lies outside its visible boundary. Failures can propagate across those interfaces, and common dependencies can cause several apparently independent Units to fail together. Redundancy therefore increases confidence only when failure modes are sufficiently independent and the coordinating logic is itself trustworthy.

Systems engineering and safety research provide strong counterweights to component-level confidence. NIST SP 800-160 states that trustworthy system behavior depends not only on trustworthy parts but on parts interacting in a trustworthy manner, and treats complexity, dependencies, uncertainty, verification, and system-level assurance as connected concerns. [S7] NASA's System Safety Handbook similarly argues for holistic treatment of systems and their constituent interactions, noting that increasing complexity can exceed the reach of traditional component-level hazard methods and that scenario analysis must address interactions, dependencies, and combinatorial effects. [S8] These sources do not prove Hero's Accountability allocation, but they support the proposition that qualified components do not automatically produce a qualified composition.

Confidence in a Composition

Composition confidence is a justified claim about the behavior of the arrangement in its intended Context. It cannot be calculated simply by averaging component confidence or counting qualified Units. It is limited by material dependencies, interface uncertainty, correlated failure, orchestration logic, observability, and the ability to intervene. A highly capable component can lower total confidence if it introduces an opaque critical dependency or operates beyond the Hero's ability to bound material consequences.

Confidence may be inherited from component qualification only where the qualification Evidence is relevant to the role the component plays in the composition. It is reduced when Context changes, when outputs are transformed before verification, when a downstream Unit treats probabilistic output as fact, when control and approval collapse into the same actor, or when the composition has no reliable stop path. System-level Evidence is therefore necessary when material behavior emerges from interaction rather than from any component in isolation.

The eight governance conditions below are not an implementation checklist in this Canon. They are theoretical validity conditions. If Capability is undefined, fitness is unproved, authority is unbounded, operation is unobservable, failure propagation is unknown, intervention is impossible, material decisions lack independent checks, or qualification cannot be withdrawn, then the Hero lacks a reasonable basis for accepting Accountability for the composition. Canon 006 may specify how these conditions are represented and tested.

Accountability Across Shared and Opaque Capabilities

The Hero is accountable for using a composition within a Value Commitment, but does not absorb every Accountability held elsewhere. The owner of a shared Capability may hold Independent Accountability for its qualified operation, evidence, limits, and change communication. The Hero relies on that Accountability while remaining answerable for selection, Context fitness, interface design, monitoring, and response. Explicitly separated Accountabilities allow scale; vague shared responsibility hides the point at which a challenge must be answered.

Opacity is not automatically disqualifying. People routinely rely on systems they cannot fully inspect, and no Hero can understand every internal mechanism in a complex Value Creation System. The relevant question is whether opacity has been compensated by sufficient external Evidence, bounded permissions, contractual or institutional assurances, independent evaluation, runtime signals, containment, escalation, and replaceability. Where a material dependency cannot be observed, bounded, challenged, or replaced, the Hero cannot responsibly convert ignorance into confidence.

This is also the answer to the counterclaim that complexity exceeds one person's understanding. Hero Accountability does not require a complete mental simulation of the system. It requires the Judgment to establish appropriate governance, obtain specialist and independent assurance, recognize the limits of available Evidence, and refuse, redefine, pause, or escalate a Commitment when the composition cannot be responsibly governed. Scale is legitimate when Accountability architecture grows with Capability architecture.

Claims to Establish

- Capabilities can be combined into a value-producing arrangement.
- Composition introduces interfaces, dependencies, failure propagation, and evidence needs.
- The Hero remains accountable for selecting and governing the composition used for a Value Commitment.

Required Proof

- Define composition, orchestration, delegation, and governance.
- Identify minimum conditions for valid composition.
- Explain how capability confidence is inherited, limited, or reduced across a chain.

Confirmed Capability Composition Governance Baseline

Capability Composition is valid only when the Hero has a reasonable basis for trusting, observing, limiting, and changing the composition used for the Value Commitment.

Minimum governance requires:

- **Capability Definition**: the expected result and operating conditions are explicit;
- **Qualification**: each Capability Unit has Evidence of fitness for the intended Context;
- **Scope and Authority**: permitted decisions, access, data, tools, and actions are bounded;
- **Observability**: execution, Outcome signals, exceptions, and relevant Evidence can be inspected;
- **Failure Boundaries**: known limits, dependencies, material failure modes, and propagation risks are understood;
- **Escalation and Stop Conditions**: the Hero can pause, replace, contain, or escalate when limits are exceeded;
- **Separation of Duties**: independent checks exist where one actor or unit must not both create and approve a material decision;
- **Revalidation and Revocation**: qualification can expire, be challenged, be restricted, or be withdrawn when Context or Evidence changes.

The Hero remains accountable for selecting the composition, judging its fitness for the Value Commitment, governing its interfaces, and responding to Evidence. A person or institution responsible for a shared Capability may hold separate Independent Accountability for the Capability's qualified operation. These accountabilities must be explicit rather than merged into general shared responsibility.

An opaque Capability Unit may be used only when sufficient external Evidence, limits, controls, and escalation paths make its use responsibly governable. If a material dependency cannot be observed, bounded, challenged, or replaced, the composition does not yet have an adequate Accountability basis.

Composition confidence must account for its weakest material dependencies, shared dependencies, and correlated failures. Adding more Capability Units does not automatically increase reliable capability.

Counterexample to Address

Complex systems may exceed any one person's ability to understand or directly control.

Boundary

Accountability does not require omniscience. It requires justified governance, escalation, evidence, and recognition of limits.

Falsification Test

The model must be revised if composed capability can create material value reliably without any accountable selection, governance, or escalation function.

7. Value Commitment

KEY IDEA

A Value Commitment converts intent into an accountable promise that can be evaluated.

Purpose

Define the commitment that connects human accountability to capability and outcome.

Theory

Intent becomes operationally meaningful only when someone becomes answerable for what the intent is expected to improve, for whom, under which boundaries, and how the claim will be evaluated. A **Value Commitment** is that conversion. It is an accountable promise to pursue and evaluate a legitimate material improvement, not a prediction that success is certain and not a guarantee that uncertainty has been removed.

This distinction protects both ambition and honesty. If Commitment meant guaranteed success, only trivial or already-known work could be accepted. If it meant nothing more than sincere effort, failure could always be excused by activity. A Value Commitment therefore binds the Hero to the quality of Judgment, Capability governance, Evidence, response, and Learning within an accepted boundary. It does not make the Hero omnipotent or make every Outcome controllable.

Value Intent precedes Capability Architecture because the question of what should improve must govern the selection of what can act. Starting with available tools and searching for a justification reverses this order. It encourages Capability utilization rather than Value creation and makes success whatever the chosen tool happens to produce. The Commitment provides the reason against which Capability is selected, composed, limited, changed, or rejected.

Commitment and Adjacent Concepts

A Value Commitment is related to, but not interchangeable with, a goal, task, requirement, or performance target.

- A **Goal** names a desired direction or state but may omit the beneficiary, legitimacy basis, Accountability center, and evaluation logic.
- A **Task** names work to be performed but does not establish why completion should matter.
- A **Requirement** constrains what a solution or process must satisfy but may not express the complete Value claim.

- A **Performance Target** sets a measured threshold but may represent only one indicator and can be achieved without the intended Outcome.
- A **Value Commitment** integrates purpose, beneficiary, material improvement, legitimacy, economic or resource logic, boundaries, Accountability, and Evidence into one answerable claim.

Goals, tasks, requirements, and targets may all sit inside a Value Commitment. None automatically becomes one. A team can complete every assigned task and meet every local target while the beneficiary experiences no material improvement or an externalized party carries an unacceptable Consequence.

Minimum Clarity Under Uncertainty

Commitment requires sufficient clarity for answerability, not false precision. At acceptance, the Hero must be able to state at least:

1. the legitimate beneficiary class and intended material improvement;
2. the purpose and institutional basis that make the Value claim legitimate;
3. the economic or resource logic through which the Commitment remains sustainable;
4. the material stakeholders, constraints, assumptions, and possible negative Consequences already visible;
5. the human Primary Accountability center and any relevant Independent Accountability boundaries;
6. the available Authority, resources, decision rights, access, governance support, and escalation paths;
7. the Baseline, intended Outcome, Evaluation Horizon, Leading and Lagging Evidence, and disconfirming Evidence;
8. the conditions under which the Commitment must be refused, changed, paused, escalated, or concluded.

These elements do not need equal detail in every Context. Their rigor should be proportionate to uncertainty, reversibility, scale, and material Consequence. The UK Magenta Book similarly treats evaluation as something that should shape an intervention before, during, and after implementation. It calls for early examination of how an intervention is expected to work, why it might not work, where uncertainty lies, which Baseline will be used, and what unintended Consequences appear. [S9] Hero uses this as supporting evidence for evaluation-aware Commitment, not as a prescribed public-policy method.

Discovery Commitments

Discovery work is not an exception to Value Commitment. It changes the object of the Commitment. When neither the solution nor the final Value path is yet knowable, the Hero may commit to reducing a material uncertainty, testing a causal assumption, identifying a viable beneficiary need, or deciding whether a larger Commitment should be made.

A valid Discovery Commitment states what uncertainty matters, why reducing it has Value, what Evidence will be gathered, which alternatives remain open, what decision the discovery is intended to enable, and when continued exploration would no longer be justified. Its Outcome may be a supported direction, a rejected hypothesis, a narrowed option set, or a responsible decision not to proceed. Producing research artifacts without changing the decision basis is activity, not completed discovery Value.

This boundary prevents two opposite failures. Premature precision forces unknown work into fictional certainty. Unlimited discovery allows uncertainty to become a permanent shelter from Accountability. A Discovery Commitment is answerable for the quality and consequence of uncertainty reduction even when it cannot promise the eventual solution.

Authority, Acceptance, and Change

Authority is part of the acceptance basis. It includes not only formal decision rights but also practical access to information, resources, Capability, governance forums, and escalation. A Hero who lacks the means to make or influence the material judgments within scope cannot responsibly accept final answerability for them.

The required response is refuse, redefine, or escalate. This rule applies before acceptance and whenever a material Authority gap becomes visible. An unforeseeable later change in Authority does not retroactively make the original acceptance irresponsible, but it creates a new decision point. The Hero must disclose the gap and seek restoration, scope change, transfer, pause, or escalation before continuing as though the Accountability basis were unchanged.

A Commitment is therefore durable but not immutable. It may change when the beneficiary, Context, Evidence, Authority, risk, Capability, economic basis, or evaluation horizon materially changes. A material change requires renewed Judgment and explicit re-acceptance; otherwise the organization quietly evaluates one promise while executing another. The original Baseline, rationale, and Evidence must remain visible enough to explain why the Commitment changed.

Commitments can also conflict. Two legitimate Commitments may compete for the same resource, impose incompatible Consequences, or sit under different Independent Accountability constraints. Conflict cannot be solved by declaring both priorities. It requires an explicit decision about scope, sequence, resource, Authority, or higher-order Value, while preserving constraints that are not legitimately tradable.

A Commitment concludes when its evaluation basis has been satisfied, when it is explicitly superseded, when its horizon or conditions expire, or when Evidence shows that the intended Value is no longer legitimate or achievable within the accepted boundary. Refusal and responsible termination are valid Accountability outcomes. Continuing an invalid Commitment merely to preserve an appearance of consistency is not perseverance; it is avoidance of Judgment.

Claims to Establish

- Value Intent precedes Capability Architecture.
- A Value Commitment names the beneficiary class, intended material improvement, legitimacy source, economic or resource logic, relevant stakeholders, constraints, and the accountable Hero.
- Commitment is not a guarantee of success; it is an answerable promise to pursue and evaluate defined value.

Confirmed Commitment-Authority Rule

A Hero must not accept a Value Commitment and later use insufficient authority as a reason to escape Accountability.

Before accepting the commitment, the Hero must determine whether the available decision rights, access, resources, governance support, and escalation paths are sufficient for responsible judgment.

If they are insufficient, the Hero must do one of three things:

1. refuse the commitment;
2. redefine its scope, authority, or conditions;
3. escalate it before acceptance or before the unresolved authority gap creates material risk.

Once a Value Commitment is accepted, a known or reasonably discoverable authority gap is not an excuse. It is evidence that the commitment was accepted without an adequate accountability basis.

Confirmed Evidence Commitment Rule

At acceptance, a Value Commitment must state enough evaluation logic to distinguish intended Value from evidenced Value. The theory requires:

- a relevant Baseline;
- the intended Outcome;
- Leading Evidence that may show credible progress;
- Lagging Evidence that may verify realized Value;
- possible negative or externalized Consequences;
- an Evaluation Horizon;
- evidence that would challenge or disprove the Value claim.

Canon 002 defines why these elements are necessary. Canon 006 may later specify their formal representation and lifecycle.

Required Proof

- Define sufficient clarity for value without pretending uncertainty can be removed.
- Distinguish commitments from goals, tasks, requirements, and performance targets.
- Explain how commitments change, expire, conflict, or are refused.

Counterexample to Address

Discovery work may begin before the value or solution can be defined precisely.

Boundary

A Value Commitment may commit to reducing uncertainty or discovering a viable value path, provided its evidence and decision conditions are explicit.

Falsification Test

The concept fails if commitments cannot be distinguished in practice from ordinary goals or assigned work.

8. Outcome, Evidence, and Consequence

KEY IDEA

Value becomes credible only when outcomes are interpreted through evidence and material consequences.

Purpose

Prevent Hero from rewarding output volume, activity, or persuasive narrative in place of value.

Theory

Value evaluation fails when several different objects are compressed into one word. **Output** is what execution produces. **Outcome** is a change in the condition or behavior of a beneficiary or system. **Value** is the Judgment that the Outcome constitutes a legitimate material improvement when its Consequences are considered. **Evidence** is what supports or challenges that Judgment. **Consequence** is an effect of the Commitment, whether intended or unintended, beneficial or harmful, direct or indirect.

The distinctions are causal as well as linguistic. An Output may contribute to an Outcome but does not prove one. An Outcome may occur without being caused by the Commitment. A measured change may benefit one stakeholder while harming another. A persuasive narrative may connect them after the fact without reliable Evidence. Hero therefore requires an explicit chain from action to Output, from Output to Outcome, and from Outcome to Value, with assumptions and alternative explanations visible.

The Magenta Book makes a comparable distinction between monitoring or benefits management and evaluation. It describes evaluation as examining whether an intervention worked, how and why it worked, what additional or unintended impacts occurred, what would have happened without it, and how far observed change can be attributed or connected to the intervention. [S9] This supports Hero's refusal to treat performance monitoring alone as verified Value.

Evidence as a Relationship

Evidence is not a privileged class of document. The same record may be strong Evidence for one claim, weak Evidence for another, and irrelevant to a third. Evidence exists in a relationship among an observation, a claim, a Context, and a Judgment. Provenance establishes where the observation came from. Relevance explains the relationship to the claim. Reliability concerns how far it can be trusted. Time Boundary establishes when it applies. Limitations identify what it cannot establish. Challengeability keeps conflicting data and alternative interpretation capable of changing the conclusion.

Evidence sufficiency is also relational. It depends on the materiality of the Consequence, the uncertainty of the causal claim, the reversibility of the decision, the cost of error, and the availability of stronger Evidence. A low-risk reversible decision may responsibly proceed on limited Evidence. A high-impact or irreversible Commitment requires greater independence, coverage, reliability, and scrutiny. Canon 002 does not set numeric thresholds; it establishes why the burden of Evidence must rise with the burden of consequence.

No individual measure should be mistaken for the complete Value claim. Quantity can hide quality, averages can hide distribution, short horizons can hide delayed harm, and success criteria can shape behavior toward the measure rather than the purpose. Once an indicator becomes a strong target, actors and systems may optimize the visible signal while degrading the underlying Outcome. The response is not to abandon measurement but to preserve the causal explanation, use multiple forms of Evidence where material, inspect distribution and negative Consequences, and keep disconfirming Evidence institutionally possible.

Conflicting Evidence is not noise to be averaged away automatically. Conflict may reveal different Contexts, stakeholder experiences, time periods, data-generation methods, or causal mechanisms. The Hero must explain whether the conflict changes confidence, narrows the claim, requires new Evidence, or demonstrates that the Value is distributed unevenly. A conclusion that cannot survive visible counterevidence is not Evidence-based; it is Evidence-decorated.

Causality, Contribution, and Delayed Outcomes

Value frequently appears after execution and may be affected by many causes. Hero does not require perfect causal attribution, but it requires precision about what is known. Where a credible counterfactual or comparison exists, the Hero may make a stronger attribution claim. Where causality is diffuse, the more honest claim may concern contribution: the Commitment plausibly influenced the Outcome through an evidenced mechanism, but did not uniquely cause it.

The causal chain should identify assumptions, external influences, and expected time lags before success is declared. S9's Theory of Change guidance similarly asks evaluators to make the causal chain, actors, affected groups, required conditions, assumptions, Evidence strength, and wider Context explicit. It also distinguishes early or medium-term outcomes from long-term impacts. [S9]

The epistemic terms Intended Value, Evidenced Progress, Provisional Value, Verified Value, and Unresolved or Refuted protect this time dimension. They prevent the organization from calling an intention a result or treating a leading indicator as final verification. Leading Evidence can justify continued investment or confidence in direction; Lagging Evidence may later establish whether the intended Outcome persisted and whether delayed Consequences changed the conclusion.

Proxy indicators are valid only when the relationship to the intended Outcome is explained and remains challengeable. Training completion may be a proxy for exposure, not changed practice. Feature use may be a proxy for engagement, not beneficiary improvement. Risk-control installation may be a proxy for protection, not proof that loss has been reduced. A proxy that improves while the intended Outcome stagnates is Evidence against the assumed causal link, not a reason to redefine Value around the proxy.

When the Evaluation Horizon extends beyond the Hero's role, tenure, or the delivery organization, Accountability does not disappear at handover. The accepting Hero must establish future review ownership, preserve the Baseline and evaluation logic, transfer relevant Evidence and limitations, and ensure that later Judgment can still challenge the original claim. The future reviewer may hold a new Accountability for evaluation; the original decision remains traceable rather than ownerless.

Material Consequence

Consequence completes Value evaluation because benefits and burdens are rarely contained inside the intended beneficiary. Consequences may be intended or unintended, immediate or delayed, concentrated or distributed, reversible or irreversible, and internalized or externalized. A Commitment that improves one metric by moving cost, risk, labor, delay, or harm outside the measured boundary has not established Value merely because the local Outcome improved.

Materiality is bounded by accepted scope, legitimate Authority, reasonable foreseeability, practical influence, and the Evaluation Horizon. These boundaries prevent infinite Accountability, but they are not loopholes. Scope cannot be drawn so narrowly that a known affected party disappears. Foreseeability requires reasonable inquiry, not only what the Hero happened to notice. Practical influence includes the ability to stop, redesign, disclose, or escalate, not only direct control.

Truly unforeseeable events and decisions governed by another explicit Independent Accountability center may lie outside the Hero's final answerability. Even then, the Hero remains answerable for whether known limits were disclosed, interfaces were governed, Evidence was monitored, and escalation occurred when the boundary became visible. Accountability is for responsible Judgment under bounded uncertainty, not retrospective ownership of everything that happened.

Some important Outcomes will remain partly unobservable. In those cases, Hero permits triangulation, qualitative Evidence, expert Judgment, scenarios, contribution analysis, and explicit uncertainty. It does not permit certainty theater. The correct conclusion may remain Provisional, Unresolved, or Refuted. A framework that forces every important Value into immediate numerical verification would systematically prefer the measurable over the meaningful and would fail its own falsification test.

Claims to Establish

- Output, outcome, value, evidence, and consequence are different concepts.
- Evidence supports judgment but does not eliminate interpretation.
- Consequences include intended, unintended, delayed, distributed, and externalized effects.

Confirmed Evidence Baseline

Evidence is traceable information, observation, or record capable of supporting or challenging a judgment about whether a Value Commitment produced its intended Outcome, material Consequences, or Learning.

Evidence must be able to change the judgment. If information cannot affect the conclusion regardless of what it shows, it is a record or decoration, not meaningful Evidence.

Minimum Evidence qualities are:

- **Provenance**: where it came from and how it was produced;
- **Relevance**: how it relates to the Value claim;
- **Reliability**: how far the observation or record can be trusted;
- **Time Boundary**: which period it represents;
- **Limitations**: what it cannot establish and where bias may exist;

- **Challengeability**: whether counterevidence, conflict, and alternative interpretation can change the judgment.

An indicator is not Evidence merely because it is measured. Its relationship to the Value claim must be explained. Output Evidence does not automatically establish Outcome or Value.

Evidence supports judgment; it does not replace judgment.

Confirmed Delayed-Outcome Baseline

When an Outcome is delayed or cannot yet be observed directly, the Value claim must retain an explicit epistemic status:

- **Intended Value**: the value intent exists, but outcome evidence does not;
- **Evidenced Progress**: Evidence supports the direction, but the intended Outcome has not yet appeared;
- **Provisional Value**: an initial Outcome is visible, but the Evaluation Horizon is incomplete;
- **Verified Value**: Evidence satisfies the declared evaluation basis;
- **Unresolved or Refuted**: Evidence is insufficient, contradictory, or disproves the Value claim.

These are terms for describing what is currently known. They are not a prescribed HeroOS workflow or lifecycle.

Delayed Value is not absent Value, but it remains unverified Value until the required Evidence arrives.

Proxy indicators may be used only when their relationship to the intended Outcome and their limitations are explicit. Improvement in a proxy must not be presented as verified Value.

A Hero must state not only what would confirm Value, but also what would disprove it.

Confirmed Material Consequence Baseline

A **material Consequence** is an effect caused or materially influenced by a Value Commitment that significantly affects a person, an organization, or society and falls within the Hero's accepted scope, legitimate authority, reasonable foreseeability, practical influence, and stated evaluation horizon.

A Hero is accountable for responsible judgment, not for omnipotence.

The Hero remains accountable for:

- defining Value and its boundaries;
- making or organizing material value judgments;
- selecting and governing Capability with reasonable care;
- monitoring available Evidence;
- stopping, refusing, redefining, or escalating when risk or authority exceeds the accepted boundary;
- explaining decisions, known trade-offs, and resulting Learning.

The Hero is not automatically accountable for genuinely unforeseeable events, matters clearly outside the accepted scope and practical influence, or decisions governed by another explicit Independent Accountability center. This exclusion applies only when the Hero exercised reasonable judgment, disclosed known limits, and used the required escalation path.

Required Proof

- Define evidence quality, provenance, relevance, timeliness, and sufficiency.
- Explain evaluation when outcomes are delayed or partly unobservable.
- Address conflicting indicators and Goodhart-style measurement failure.

Counterexample to Address

Some high-value decisions cannot be evaluated within the decision-maker's tenure or through direct causal attribution.

Boundary

Hero requires explicit evidence logic, not perfect measurement or immediate certainty.

Falsification Test

The framework must be revised if its evidence requirements systematically favor short-term measurable outputs over important long-term value.

9. Learning and Organizational Memory

KEY IDEA

Learning exists only when evidence changes future judgment, behavior, memory, or capability.

Purpose

Explain why execution should make the organization more capable rather than merely create records.

Theory

Learning is the change produced when Evidence alters future Judgment, behavior, or Capability. Information can be received without being understood. Insight can occur without being retained. A lesson can be documented without being used. Hero therefore distinguishes `recording`, `individual learning`, `organizational learning`, and `Organizational Memory`.

Recording preserves information. Individual learning changes what one person understands or does. Organizational learning changes a shared practice, governance choice, Capability, or decision pattern. Organizational Memory makes that learning durable and recoverable beyond the continued presence or recollection of the original people. The four can occur together, but none should be inferred from storage alone.

NASA's Knowledge Policy for Programs and Projects provides a useful institutional example. It treats capture, retention, utilization, and sharing as distinct activities; places its Lessons Learned Information System inside a wider system of knowledge transfer, development, events, and tools; and calls for lessons to be infused into methodologies, standards, training, and practice. [S10] This does not prove Hero's four-part definition, but it supports the distinction between possessing a repository and changing organizational operation.

From Evidence to Memory

Organizational Memory requires a traceable transformation:

```
Retained Evidence
+ Preserved Context
+ Explicit Judgment
+ Operational Change
= Organizational Memory
```

Retained Evidence preserves what was observed, including conflicting or negative results where material. **Preserved Context** records the conditions under which the Evidence and Judgment were relevant: purpose, stakeholders, Baseline, Authority, Capability, assumptions, time, constraints, and environment. **Explicit Judgment** states what was concluded, with uncertainty, alternatives, and limitations. **Operational Change** embeds the conclusion into what the organization will later do, such as a decision rule, Capability qualification, design principle, control, training practice, monitoring question, workflow, tool, or escalation condition.

The formula is not a database schema. It is a validity test. Evidence without Context is easily misapplied. Context and Evidence without Judgment force every future user to reconstruct the decision. Judgment without Evidence becomes unsupported doctrine. All three without Operational Change create an archive of lessons that the organization is free to repeat.

Memory becomes useful through retention, retrieval, interpretation, and application. Retention keeps the material available and authentic. Retrieval brings the relevant material into a future decision. Interpretation checks whether the old Context fits the new Context. Application changes current Judgment or action. Failure at any stage breaks the memory function. Material that cannot be found is not operational memory; material retrieved without Context may be more dangerous than material forgotten.

Context fit is therefore mandatory. Learning changes Context(n) into Context(n+1), but the resulting Judgment is not universally reusable. A practice learned during crisis, under a particular regulation, with a specific Capability, or for one beneficiary class may fail elsewhere. Reuse requires asking which causal conditions remain, which have changed, and whether new Evidence challenges the old conclusion.

Tacit Learning and Durability

Organizations sometimes change behavior without preserving explicit Evidence. Routines, stories, supervision, culture, and skilled imitation can carry tacit learning. Hero recognizes this as real organizational learning when the changed behavior is detectable. It does not automatically call it durable Organizational Memory.

Tacit transmission has strengths: it can preserve judgment that is difficult to formalize and can adapt through practice. It is also fragile. It may disappear with turnover, mutate without challenge, retain obsolete assumptions, or become a rule whose origin and boundary no one can explain. Where material Consequences are involved, the organization should make enough Evidence, Context, and Judgment explicit to support challenge and responsible reuse. The purpose is not to document every intuition; it is to prevent critical learning from depending on untraceable inheritance.

Exit, Supersession, and Capacity

Memory is not valuable merely because it persists. An organization that can learn but cannot unlearn accumulates conflicting rules, obsolete assumptions, harmful bias, and excessive retrieval noise. Organizational Memory therefore needs an exit mechanism as much as an admission mechanism.

Active, Conditional, Retired or Superseded, Archived, and Disposed describe different relationships to present Judgment. They are conceptual distinctions, not Canon 006 lifecycle states. **Active Memory** influences decisions by default. **Conditional Memory** applies only when its Context conditions are met. **Retired or Superseded Memory** no longer has current decision authority. **Archived Evidence** remains available for audit, research, or historical reconstruction. **Disposed Material** is deleted when law, privacy, security, retention, or legitimate resource constraints require it.

Records-management practice provides an external analogy but not an identity. NARA guidance distinguishes temporary records that should be destroyed after approved retention, permanent records that must be preserved and transferred, superseded authorities, and schedules that require review and updating. It also warns that vague instructions such as “destroy when no longer needed” can lead to over-retention or premature destruction. [S11] Organizational Memory is broader than federal records management, but the example supports explicit disposition, authority, review, and separation between current use and historical preservation.

Retirement is real only when decision influence changes. A memory marked obsolete but still returned by default search, AI retrieval, qualification rules, training, controls, or workflow remains active in practice. Supersession must preserve traceability while directing current Judgment to the newer conclusion. Disposal must remove material lawfully without pretending that deletion proves the organization has unlearned the behavior it once embedded.

Capacity limits apply to active influence, not to all history. Historical Evidence may remain extensive when lawful, secure, useful, and affordable. Active Memory competes for organizational attention and can create contradiction, delay, or automatic bias. The proper objective is therefore the smallest sufficient active set for responsible Judgment, not a universal maximum item count. What is sufficient depends on domain complexity, consequence, rate of change, retrieval quality, and the organization's ability to resolve conflict.

Each Active Memory item must have enough ownership and boundary to remain challengeable: an applicable Context, supporting Evidence, explicit Judgment, operational effect, effective date, review or expiry condition, and known dependency or conflict. Canon 006 may determine how these are represented, reviewed, counted, or automated. Canon 002 establishes why memory without ownership, boundary, and exit cannot remain trustworthy.

When an Outcome will mature after a Hero's tenure, Organizational Memory carries the evaluation forward. The Baseline, causal assumptions, Evidence status, limitations, review horizon, and future owner must survive the handover. Without that continuity, long-term Value becomes structurally unverifiable and organizations are rewarded for declaring benefit before Consequence can appear.

Claims to Establish

- Stored information is not automatically Organizational Memory.
- Learning requires a detectable change in future action, judgment, or available capability.
- Memory must preserve both evidence and the context needed to interpret it.

Confirmed Organizational Memory Baseline

Organizational Memory is retained Evidence, Context, and Judgment that has been embedded into future behavior, governance, or Capability so that Learning does not depend on one person remaining in the organization.

```
Retained Evidence
+ Preserved Context
+ Explicit Judgment
+ Operational Change
= Organizational Memory
```

- Evidence records what was observed.
- Context preserves the conditions under which the Evidence and Judgment were valid.
- Judgment records what was concluded, including uncertainty and rejected alternatives.
- Operational Change embeds the Learning into future decisions, rules, Capability qualification, workflows, controls, tools, training, or monitoring.

Stored material without Operational Change is an archive or knowledge repository, not Organizational Memory. A rule without supporting Evidence and Context is an unsupported rule. Knowledge held only by one person is fragile tacit knowledge, not organizational memory.

Memory without Context becomes superstition.

Confirmed Memory Exit and Capacity Rule

A learning organization must also be able to unlearn. Organizational Memory must support admission, contextual use, review, restriction, supersession, retirement, archival, and lawful disposal.

Conceptually, memory may be:

- **Active Memory**: currently and by default influences judgment, governance, or Capability;
- **Conditional Memory**: applies only when defined Context conditions are present;
- **Retired or Superseded Memory**: must no longer guide current decisions;
- **Archived Evidence**: retained for audit, research, or historical traceability without current decision authority;
- **Disposed Material**: deleted when privacy, law, security, or retention obligations require it.

Retirement must remove the memory from active decision influence, including default search, AI retrieval and context, decision rules, Capability qualification, workflows, controls, and training. Memory marked retired but still retrieved as current has not actually exited.

Memory should be reviewed, restricted, superseded, retired, or disposed when Context changes, stronger counterevidence appears, related Capability or law changes, validity expires, duplication or conflict grows, harmful bias emerges, the Value Commitment ends, or lawful retention requires disposal.

The active memory set should be the smallest sufficient set required for responsible future judgment. Canon 002 does not impose a universal item count. It requires every Active Memory item to have an owner, applicable Context, supporting Evidence, explicit Judgment, operational effect, effective date, review or expiry condition, and known dependencies or conflicts.

Historical Evidence may remain extensive when lawful and useful. The constrained resource is active organizational attention and decision influence.

Memory without an exit mechanism becomes institutional inertia.

Required Proof

- Define retention, retrieval, interpretation, and behavior change.
- Explain how obsolete or harmful memory is challenged and retired.

If the Evaluation Horizon extends beyond the Hero's role or tenure, the Hero remains responsible for establishing the Evidence handover, future review ownership, and Organizational Memory needed to continue evaluation.

Counterexample to Address

An organization may learn tacitly through changed behavior without retaining explicit evidence.

Boundary

Organizational Memory does not require every retained record to remain active. It requires a traceable relationship between retained Evidence and Context, explicit Judgment, and a change embedded in future operation. Formal lifecycle states, quotas, retention periods, and automation remain Canon 006 concerns.

Falsification Test

The claim must be narrowed if useful organizational learning repeatedly occurs without retained evidence, changed behavior, changed judgment, or changed capability.

10. The Hero Model

KEY IDEA

The Hero Model describes a complete value-creation cycle: Purpose, Responsibility, Capability, Value, and Learning.

Purpose

Derive the relationship among the five elements without turning the model into a rigid workflow.

Theory

The Hero Model is a conceptual completeness test for accountable value creation. It asks whether five necessary functions are present: Purpose, Responsibility, Capability, Value, and Learning. The functions form a cycle because each changes the conditions under which the others are interpreted, but the cycle is not a mandatory sequence of meetings, phases, or hand-offs.

The model is intentionally smaller than a complete operating system. It does not describe every activity needed to deliver a product, service, policy, or mission. It identifies the minimum conceptual functions that prevent execution from becoming detached from meaning, human obligation, fitness, consequence, and adaptation.

Why the Five Elements Cannot Substitute for One Another

Purpose establishes what matters and why. It identifies the direction, legitimate beneficiary, institutional basis, and relevant horizon against which action can be judged. Purpose without Responsibility remains aspiration. It can inspire activity but cannot identify who is obliged to act or ensure action occurs.

Responsibility establishes the duty to act, perform, contribute, or ensure that work occurs. It can be distributed, delegated, and shared across participants. Responsibility without Capability is an obligation without a credible means of performance. It may create pressure, but pressure is not execution capacity.

Capability makes pursuit possible. It supplies the qualified ability to produce a result under stated conditions, whether through people, software, AI, services, platforms, workflows, or institutions. Capability without Value is power without a legitimate test of whether its use mattered. It can generate Output, scale, and efficiency while moving away from Purpose.

Value evaluates whether the Outcome was a legitimate material improvement when Evidence and Consequence are considered. Value without Learning leaves the evaluation isolated. The organization may recognize success or failure yet enter the next Commitment with the same assumptions, qualifications, controls, and blind spots.

Learning changes future Judgment, behavior, Capability, governance, or Context. Learning without renewed Purpose can optimize yesterday's priorities after they have ceased to be legitimate. Learning therefore closes one cycle only by changing the starting conditions of the next.

No element is a synonym for another. Purpose cannot execute. Responsibility cannot create fitness. Capability cannot define legitimacy. Value cannot preserve itself. Learning cannot determine what should matter forever. The model is complete only when these different functions remain connected without being collapsed.

Accountability Across the Cycle

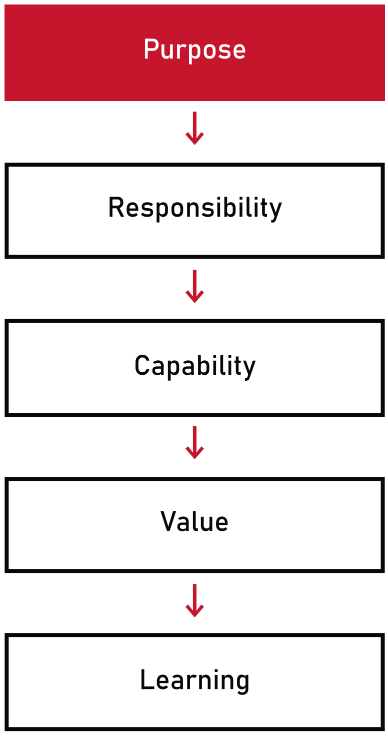
Accountability is not a sixth element because it is not one more type of work performed after Learning. It is the bounded final answerability that governs the Hero's Judgment across the complete cycle: how Purpose was interpreted, how Responsibility was organized, why Capability was considered fit, how Value and Consequence were evaluated, and what Learning changed.

Responsibility may move through the system while Accountability remains traceable. A Hero may delegate execution, rely on Independent Accountability, or use shared Capability without becoming responsible for every local action. The Hero nevertheless remains answerable within the Value Commitment for selection, integration, response to Evidence, and escalation. Accountability is therefore a relationship surrounding decisions, not a task inside the task list.

Context, Feedback, and Non-Sequential Work

Context surrounds the Hero Model because every element changes meaning with conditions. Purpose depends on legitimate beneficiaries and institutions. Responsibility depends on Authority and role boundaries. Capability depends on operating conditions. Value depends on Baseline, Evidence, time, and Consequence. Learning depends on which causal conditions remain valid.

The relationship can be expressed as:



The arrows express dependency and feedback, not required chronology. Work may begin with a crisis, available Capability, unexpected Evidence, a new legal constraint, or inherited Organizational Memory. A crisis can impose immediate Responsibility before Purpose is fully articulated. A new Capability can reveal previously impossible Value. Evidence can invalidate a Commitment before planned execution completes. Learning can force Purpose to be reconsidered rather than merely beginning another delivery cycle.

In each case, the model asks which functions are already present, which remain implicit, and which must be made answerable before the work can responsibly continue. Interruption, iteration, parallelism, and revision are therefore native to the model. The failure is not entering at the "wrong" step; it is allowing a necessary function to disappear because work entered elsewhere.

NASA's Systems Engineering Handbook provides a useful systems counterexample to linear interpretation. It describes engineering processes as recursive and iterative across a lifecycle and distinguishes verification of requirements from validation that a product serves its intended purpose and stakeholder expectations. [S12] Hero does not import NASA's process architecture, but the distinction reinforces two propositions: a conceptual cycle need not be a linear workflow, and conforming Output is not identical to validated Value.

Feedback may operate at several speeds. Immediate Evidence may change Capability use during execution. Outcome Evidence may revise the Value Judgment after delivery. Long-horizon Consequence may alter Organizational Memory much later. The Hero Model must keep these loops connected so that fast execution cannot outrun slower evaluation and Learning.

The HERO Principles remain a public mnemonic for remembering related principles. They cannot replace the Hero Model because a mnemonic does not itself establish the complete relationship among Purpose, Responsibility, Capability, Value, Learning, Accountability, and Context. The two artifacts serve different functions and should not be forced into mutual substitution.

Claims to Establish

- Purpose defines what matters.
- Responsibility establishes the human obligation to act, perform, or ensure that work occurs.

- Capability makes pursuit possible.
- Value evaluates whether the pursuit mattered.
- Learning changes the next cycle.

Accountability is not a sixth sequential step. It is the answerability relationship that governs judgment, value, and material consequence across the complete cycle.

Required Proof

- Explain why none of the five elements can substitute for another.
- Define feedback and the role of Context.
- Show how the model supports iteration, interruption, and revision.

Confirmed Context Baseline

Context is the set of conditions that determines how Purpose, Responsibility, Capability, Value, Evidence, Consequence, and Learning are interpreted.

Context includes beneficiaries and Stakeholders, Baseline, Authority and Accountability boundaries, time and place, resource conditions, legal and ethical constraints, risk, assumptions and uncertainty, Capability operating conditions, the Evidence environment, and relevant historical Judgment or Organizational Memory.

Context is not a sixth sequential step. It is the cross-cycle interpretive boundary surrounding the complete Hero Model.

```
Context(n)
    |
Hero Model Cycle
    |
Learning
    |
Context(n+1)
```

Learning changes the Context in which the next cycle is judged. No judgment is valid outside the Context in which its Evidence was produced. Reuse therefore requires a Context-fit check rather than direct repetition of an earlier conclusion.

Counterexample to Address

Real work rarely follows a clean sequence and may begin with evidence, crisis, or an available capability.

Boundary

The Hero Model is a conceptual completeness test, not a mandated execution sequence. The HERO Principles remain a public mnemonic and do not replace it.

Falsification Test

The model must be revised if a recurring class of accountable value creation requires an essential element not represented by the five-part cycle or Context.

11. One Hero Team

KEY IDEA

One Hero Team is one accountable human center surrounded by governed Capability Units.

Purpose

Explain the smallest useful organizational expression of Hero without declaring conventional teams obsolete.

Theory

One Hero Team is an Accountability topology, not a prescribed headcount, reporting line, or employment model. Its defining center is one human Primary Accountability for one Value Commitment. Around that center is the governed Capability required to pursue, evaluate, and learn from the Commitment.

The word `one` refers to the identifiability of the Primary Accountability center, not to unilateral power or solitary work. The word `team` refers to coordinated participation in the Value Commitment, not only to people who report to the Hero. A Capability Unit may participate for one decision, one phase, one execution, or many Commitments. It may be internal or external, human or technical, dedicated or shared.

The minimum conceptual structure contains:

- a Value Commitment with a legitimate beneficiary, boundary, and evaluation basis;
- a named human Hero holding Primary Accountability within that boundary;
- distributed Responsibilities for action, performance, review, and support;
- qualified Capability Units composed for the intended Context;
- explicit Authority, Independent Accountability, and institutional constraints;
- Evidence, escalation, intervention, and stop paths;
- Learning and Organizational Memory that survive individual participation.

This is a theoretical structure, not an organizational template. A conventional product team may satisfy it. A temporary mission group, procurement network, public-private program, professional case team, or human-AI arrangement may also satisfy it. Conversely, a department with stable membership may not form One Hero Team if no Value Commitment and Primary Accountability can be identified.

Participation and Governance

Team membership is Capability participation. The relevant question is not only "Who belongs to this manager?" but "Which qualified Capability is relied upon for this Commitment, under what conditions, with which Responsibility, Authority, Evidence, and exit path?" This perspective makes shared services, specialists, contractors, platforms, and AI visible without pretending they have identical legal or moral status.

Qualification establishes why a Unit may be relied upon in the intended Context. **Delegation** allocates bounded work or decisions without making Primary Accountability disappear. **Coordination** governs sequence, information, interfaces, and conflict across Units. **Replacement** ensures that a Unit can be restricted or changed when fitness fails. **Escalation** connects the Team to Authority and Independent Accountability beyond the Hero's boundary.

These functions are necessary because Capability Composition changes the nature of failure. A strong Unit can be used in the wrong Context. Two qualified Units can exchange incomplete information. A shared service can fail several Teams at once. A specialist can identify a constraint the Hero lacks Authority to resolve. Team governance therefore concerns relationships and interfaces as much as individual performance.

S12 defines a system as elements functioning together to produce a required Capability and states that system-level value beyond the independent parts is primarily created by the relationships among those parts. It also treats multidisciplinary integration as balancing opposing constraints into a coherent whole rather than allowing one discipline to dominate. [S12] This supports the One Hero Team emphasis on governed relationships, while leaving Hero's Accountability model as a separate theoretical claim.

Primary and Independent Accountability

One Primary Accountability does not mean one person makes every decision. Safety, security, finance, law, professional ethics, democratic authority, audit, and other domains may require Independent Accountability with separate decision rights, approvals, vetoes, or duties. These constraints are not subordinate Capabilities that the Hero may optimize away.

The structure remains coherent when each Accountability has an explicit object and boundary. The Hero answers for the Value Commitment. A safety authority may answer for an accepted safety constraint. A data custodian may answer for lawful use. A shared Capability owner may answer for qualification and operation of the shared service. Conflict among these centers requires an authorized resolution path; declaring Accountability "shared" without boundaries makes none of them reliably answerable.

Primary Accountability also remains bounded. The Hero is not accountable for governing a shared Capability in every use across the organization. The Hero is accountable for the decision to rely on it in this Commitment, for Context fitness and interface governance, and for response when Evidence changes. The shared Capability owner remains independently answerable for the basis on which the Unit was represented as qualified.

Scale and the Value Creation System

Scale in Hero is accountable Value created through governed Capability Composition, not the number of people reporting to a manager. Adding headcount, agents, services, automation, or parallel work increases nominal execution capacity. It increases legitimate scale only when the Value Commitments, interfaces, Evidence, Authority, and failure boundaries remain governable.

The unit above One Hero Team is therefore a `Value Creation System`: a network of related Value Commitments, Hero Teams, Capability Units, governance boundaries, Evidence, and Organizational Memory. Its boundary follows Value, Accountability, Capability dependency, and Consequence. It may cross departments or sit inside one. It may be temporary or enduring. Reporting hierarchy alone cannot define it because hierarchy may not follow the flow of Value or failure.

Accountability does not automatically aggregate upward. A manager of several Heroes is not automatically a Super Hero for all their Commitments. A system-level Hero exists only when someone accepts a distinct system-level Value Commitment with sufficient Authority and an evaluation basis that is not reducible to local Commitments. This system Accountability coordinates and judges system Value; it does not erase local Primary and Independent Accountabilities.

This treatment allows nested and connected systems without demanding one ultimate human owner for every consequence in the organization. It preserves identifiable answerability at each accepted Value boundary and requires explicit governance where Commitments compete, dependencies cross boundaries, or Consequences propagate.

Claims to Establish

- Team membership can be understood as capability participation, not only employment or reporting line.
- Human specialists, AI agents, services, platforms, and workflows may all serve as Capability Units.
- Scale is accountable value created through governed Capability Composition.
- One Hero Team combines one Primary Accountability center, distributed Responsibility, necessary independent checks, and explicit institutional boundaries.

Required Proof

- Explain qualification, delegation, coordination, replacement, and escalation.
- Address separation of duties and contexts where multiple humans must hold distinct accountabilities.

Confirmed Multi-Hero System Baseline

The unit above One Hero Team is a `Value Creation System`: a network of Value Commitments, Hero Teams, Capability Units, governance boundaries, Evidence, and Organizational Memory organized around related Value.

Each Value Commitment retains one named human Primary Accountability center. Accountability does not automatically aggregate upward into a single “Super Hero.”

A system-level Hero exists only when a person explicitly accepts a distinct system-level Value Commitment with sufficient Authority, scope, Evidence, and governance support. That system-level Accountability does not erase the Primary or Independent Accountabilities inside the system.

Shared Capability Units may serve multiple Hero Teams. The accountability for maintaining a qualified shared Capability is distinct from each consuming Hero's accountability for judging whether and how that Capability is used within a specific Value Commitment.

Conflicts among Value Commitments require an explicit governance mandate, decision rights, prioritization basis, and escalation path. They must not be resolved through ambiguous shared Accountability or reporting hierarchy alone.

Value Creation Systems may be nested or connected, but their boundaries should follow Value, Accountability, Capability dependency, and Consequence rather than being assumed to equal departments.

Counterexample to Address

Safety-critical, regulated, democratic, or high-conflict domains may require multiple independent decision rights.

Boundary

One Hero Team does not mean one person has unilateral authority, executes everything, or replaces institutional governance. Independent Accountability may coexist around the same Value Commitment when its scope and decision rights remain explicit.

Falsification Test

The construct must be revised if it cannot represent domains that require independent checks without collapsing accountability into ambiguity.

12. From Hero to HeroOS

KEY IDEA

HeroOS is implied when Hero principles are made repeatable, governable, observable, and learnable at organizational scale.

Purpose

Show the derivation from philosophy to operating model while reserving formal requirements for Canon 006.

Theory

Hero begins as a theory of accountable value creation. It becomes an operating-system question when the theory must remain reliable across repeated Commitments, multiple people and institutions, composed Capability, long evaluation horizons, turnover, conflict, and organizational scale.

At very small scale, Hero principles may be embodied informally. One person can remember the Commitment, know the collaborators, inspect the work, retain the Evidence, and carry the Learning forward. This is a valid counterexample to the claim that every Hero practice requires a named platform, ceremony, or formal program. Informality is not invalid merely because it is informal.

The limitation appears when answerability depends on memory, proximity, stable personnel, or exceptional personal discipline. As the Value Creation System grows, Capability and Accountability cross boundaries. Evidence arrives after handover. Shared Units change. Independent constraints conflict. Old Learning is retrieved in new Contexts. At that point, repeatability requires mechanisms that make the theory observable and governable beyond the immediate participants.

HeroOS is the name for that operating-model implication. It is not necessarily software and is never only software. It may include social practice, institutional authority, governance, records, human routines, technical tools, interfaces, and automation. Its purpose is to preserve the relationships required by Hero theory when those relationships can no longer be held reliably by informal memory alone.

Deriving the Required Functions

The derivation proceeds from failure conditions:

- If Purpose and Value Commitment cannot be identified, Capability may be optimized without a legitimate Value claim.

- If Primary and Independent Accountability cannot be identified, delegation and hierarchy can make answerability disappear.
- If Authority and scope cannot be related to Commitment, a Hero may accept a promise that cannot be responsibly governed.
- If Capability and qualification cannot be described, access may be mistaken for fitness.
- If Composition, interfaces, and failure boundaries cannot be observed, scale may amplify hidden dependency and correlated failure.
- If Outcome, Evidence, and Consequence cannot be evaluated over time, Output can be presented as Value.
- If Learning cannot become Organizational Memory, the system depends on individual retention and repeats preventable failure.
- If Memory cannot be challenged and retired, past Learning becomes current inertia.

These failure conditions imply conceptual functions for Commitment identity, Accountability and Authority, Capability qualification and Composition governance, Evidence and Consequence evaluation, escalation and intervention, and Organizational Memory. They do not determine the formal role names, record fields, state machines, workflows, thresholds, software, or conformance tests used to realize those functions.

The NASA Systems Engineering Handbook makes a related functional observation: systems engineering functions should be performed regardless of whether a manager, engineer, internal team, or contractor performs them, while the formality and documentation should vary with type, size, and complexity. [S12] HeroOS adopts the principle of necessary function with proportionate realization, not NASA's specific roles or process set.

Proportionality and Minimality

HeroOS is justified only to the extent that it preserves answerability and Value at scale. More process is not automatically more governance. A mechanism that produces records without improving Judgment, hides Authority behind ceremony, delays intervention, or overloads active memory can undermine the theory it was intended to implement.

The theoretical requirement is therefore **minimum sufficient operating structure**. The structure must be strong enough to keep Commitment, Accountability, Capability, Evidence, Consequence, and Learning traceable and challengeable. It should be no more elaborate than the Context and material Consequence require. A low-risk, short-horizon Commitment may need little explicit structure. A regulated, safety-critical, long-horizon, or multi-institution Value Creation System may require much more.

Technology agnosticism follows from this principle. HeroOS cannot be equated with an AI platform, workflow product, project-management method, organization chart, or data repository. Any of these may implement part of the operating model. None can establish conformance by brand or installation alone.

Why Canon 006 Is Necessary

Canon 002 can prove only that certain functions and validity conditions are theoretically necessary. It cannot make HeroOS testable without specifying how an organization identifies roles and permissions, represents Commitments and Evidence, governs lifecycle and state transitions, manages interfaces and escalation, controls access and retention, and determines whether required conditions have been satisfied.

That specification task belongs to Canon 006. Only Canon 006 may turn theoretical description into formal requirements and conformance tests. Canon 002's terms such as Intended Value, Verified Value, Active Memory, Qualification, or Revocation remain descriptive until Canon 006 explicitly defines an implementation contract.

The boundary also protects experimentation. Different organizations may embody Hero theory through different structures before HeroOS is standardized. Their practice can supply Evidence for or against the Foundation. Canon 006 should not be smuggled backward into Canon 002 merely because one implementation appears useful.

The derivation fails if informal practice can preserve the complete Hero relationships reliably at organizational scale without any repeatable mechanism. The word `HeroOS` may then be unnecessary. But if Commitments, Accountabilities, Capability, Evidence, Consequence, and Learning repeatedly disappear or become ambiguous without durable mechanisms, an operating model is not optional; only its implementation form remains open.

Derivation

If value requires an identifiable accountable human, an operating model needs a way to identify a Hero and a Value Commitment.

If value can be pursued through composed capability, an operating model needs a way to describe, qualify, select, and govern Capabilities and Capability Units.

If output is insufficient, an operating model needs outcome, evidence, and consequence evaluation.

If execution should improve the organization, an operating model needs learning and Organizational Memory.

These needs imply the conceptual functions of HeroOS. They do not yet specify roles, states, ceremonies, schemas, interfaces, controls, or conformance rules.

Canon Boundary

Canon 002 may define why these functions are necessary and how they relate. Canon 006 must define the formal operating requirements and testable conformance model.

Confirmed Canon 002 / Canon 006 Boundary

Canon 002 explains `why a valid Hero theory requires a concept, relationship, boundary, or necessary condition`.

Canon 006 specifies `who performs what action, when, using which role, permission, record, state, interface, control, threshold, or conformance test`.

Canon 002 may define theoretical propositions, conceptual relationships, necessary validity conditions, counterclaims, boundaries, and falsification tests. It may state that Capability requires Qualification, Composition requires Observability and Revocation, Value requires Evidence, and Organizational Memory requires exit.

Canon 006 must define formal roles and permissions, required fields and schemas, lifecycle and state transitions, workflows and approvals, escalation and revocation procedures, retention rules, interfaces, operational metrics, thresholds, and conformance requirements.

Only Canon 006 may formally determine whether an operating implementation conforms to HeroOS. Terms such as Active Memory, Verified Value, or Intended Value in Canon 002 are theoretical description language and do not automatically become HeroOS lifecycle states.

Canon 002 must remain technology agnostic and must not prescribe organizational titles, software platforms, AI products, or implementation mechanisms.

The editorial test is:

- if a statement answers “Why is this necessary for a valid Hero theory?”, it belongs in Canon 002;
- if it answers “Who does what, when, using which operating mechanism or control?”, it belongs in Canon 006.

Counterexample to Address

An organization may embody Hero principles informally without installing a named operating model.

Falsification Test

The derivation fails if Hero's claims can be sustained at scale without any repeatable mechanism for commitment, capability governance, evidence, consequence, and learning.

13. Limits, Counterclaims, and Falsifiability

KEY IDEA

Hero is a theory only if its claims can be bounded, challenged, and revised.

What Falsifiability Means Here

Falsifiability is stronger than the permission to criticize. A theory can absorb unlimited criticism by redefining its terms after every failure. Hero must instead state what kind of Evidence would cause a proposition to be retained, narrowed, qualified, replaced, or rejected.

Not every Hero claim is testable in the same way. An empirical claim, such as the Scarcity-Shift Hypothesis, can be compared with observed changes in cost, availability, productivity, failure, and constraint. A structural claim, such as the necessity of Evidence for Verified Value, can be challenged by a counterexample that repeatedly produces warranted evaluation without the proposed relationship. A normative claim, such as keeping human beings at the center of Value legitimacy, cannot be proved by productivity data alone; it must be challenged through ethical, legal, political, and institutional reasoning as well as consequences. Failure to distinguish these claim types would make apparent precision misleading.

Canon 002 therefore does not claim that S1-S12 prove Hero. Those sources establish Context, examples, counterevidence, or adjacent disciplines. Hero's proprietary relationships remain hypotheses and arguments that require independent comparison, field Evidence, and specialist review.

Scope Limits

Hero is not a universal theory of persons, morality, government, economics, consciousness, or intelligence. It is a bounded theory of accountable Value creation under conditions in which Capability can be accessed, composed, delegated, or automated.

The theory is most relevant when work has material Consequence, multiple possible means, uncertainty about outcomes, and a need to connect Judgment with answerability and learning. It may add little where action is trivial, fully prescribed, consequence-free, or already governed by an institution that preserves the same necessary relationships under different names.

The Hero vocabulary is not evidence of Hero validity. Renaming a product owner, manager, professional, public official, or autonomous-agent supervisor as a Hero proves nothing. The theory earns distinctiveness only if its combined propositions explain failures or enable better Judgment that existing models do not already explain as well.

Counterclaim 1 - Execution May Remain the Bottleneck

Counterclaim. In many domains, intelligence-enabled Capability does not make Execution abundant. Physical work, trusted relationships, tacit expertise, regulatory approval, capital, infrastructure, attention, coordination, and local Context may remain scarce. Additional AI can even slow experienced practitioners in high-context work, as the bounded evidence in S3 and S4 shows.

Current response. Hero does not require Execution to become universally cheap. The Scarcity-Shift Hypothesis claims that the relative constraint can move in some Contexts: as access to certain forms of execution expands, choosing worthy Value, accepting bounded Accountability, composing Capability, and evaluating Consequence become more visible constraints. S1 and S2 support parts of that possibility; S3 and S4 prevent it from becoming a universal law.

Boundary and uncertainty. The shift may be absent, partial, temporary, or reversed. It may occur in one stage of a Value Creation System while another stage remains execution-constrained.

Revision trigger. Narrow or reject the scarcity-shift proposition if comparative Evidence across relevant domains shows that increased access to intelligence-enabled Capability does not materially change the location of constraint, the allocation of human Judgment, or the need for accountable selection. Hero may still survive as an accountability theory, but not as a theory grounded in a broad scarcity shift.

Counterclaim 2 - Accountability Can Be Collective or Institutional

Counterclaim. Complex outcomes are often produced and governed collectively. Boards, courts, professions, democratic bodies, incident commands, partnerships, and communities may legitimately answer as institutions. Requiring one human Primary Accountability center can oversimplify shared causation, erase independent duties, create a blame target, or grant one person authority they should not possess.

Current response. Hero distinguishes distributed Responsibility, Primary Accountability for a bounded Value Commitment, and Independent Accountability for distinct constraints. The individual center is intended to prevent final answerability from disappearing, not to claim sole causation, unilateral authority, or ownership of every consequence. Chapters 3 and 11 expressly preserve institutional and independent forms around the primary center.

Boundary and uncertainty. The present theory has not established that every legitimate Value Commitment can or should have one individual Primary Accountability center. Some constitutional, judicial, scientific, fiduciary, cooperative, or democratic decisions may be valid only when final answerability remains genuinely plural.

Revision trigger. Replace the one-human-center requirement with a broader identifiable-accountability rule if comparative institutional Evidence shows that durable collective or institutional structures can preserve bounded final answerability, challengeability, authority alignment, and learning as well as or better than an individual center, without creating ownerless commitments.

Counterclaim 3 - Governed Machine Judgment May Be More Reliable or Legitimate

Counterclaim. Human Judgment is biased, inconsistent, fatigued, corruptible, and sometimes inaccessible. A governed machine system may outperform individuals in accuracy, consistency, auditability, or equal treatment. In some narrowly authorized decisions, legitimacy may come from law, public procedure, or institutional design rather than from a particular human making the judgment.

Current response. Hero does not reserve analysis, recommendation, optimization, or even bounded decision execution to humans. It reserves the legitimacy of Value and final answerability for material Consequence within a human institution. NIST's AI RMF supports human-centric governance and responsibility across AI actors but does not prove Hero's specific allocation [S5]. Reliability and legitimacy are different: superior prediction does not by itself establish whose interests count, which harms are acceptable, or who must answer when the system is wrong.

Boundary and uncertainty. Human review can be ceremonial, less informed than the machine, or incapable of changing an outcome. A theory that merely inserts a nominal person into an automated chain would fail its own accountability standard.

Revision trigger. Narrow or replace the human-primary requirement if a governed non-human decision arrangement can demonstrably hold legitimate authority, participate in challenge and remedy, bear or reliably allocate consequence, and sustain answerability without a human accountability center. Until then, machine superiority in performance supports broader Capability delegation, not the disappearance of human institutional Accountability.

Counterclaim 4 - Value May Emerge Through Discovery

Counterclaim. Scientific research, design, entrepreneurship, diagnosis, exploration, art, and emergency response often begin before the valuable outcome is knowable. Requiring Value to be defined first can reward false certainty, suppress discovery, and convert learning into failure.

Current response. A Value Commitment need not promise a predetermined solution or verified benefit. It can commit to a bounded Purpose, beneficiary, question, learning objective, option creation, risk reduction, or decision that is valuable under stated uncertainty. Chapter 7 permits revision when Evidence changes the understanding of Value; it does not permit unlimited activity without an accountable reason to continue.

Boundary and uncertainty. Some discovery may resist even a stable beneficiary or outcome frame at the beginning. Retrospective narratives can also make purposeless exploration appear intentionally valuable.

Revision trigger. Qualify the Value Commitment requirement if field Evidence shows that valuable discovery is systematically damaged by even provisional pre-commitment. Reject it as a universal entry condition if open exploration can remain governable, challengeable, and resource-justifiable without any prior statement of Purpose, beneficiary, constraint, or learning value.

Counterclaim 5 - Evidence May Be Delayed, Political, or Incomplete

Counterclaim. Outcomes can take years, causality can be diffuse, affected people can disagree, and powerful actors can control what is measured. Evidence can create false confidence, privilege measurable interests, or arrive too late to ground meaningful Accountability.

Current response. Hero separates Intended, Evidenced Progress, Provisional, Verified, and Unresolved or Refuted Value language. It requires Provenance, Relevance, Reliability, Time Boundary, Limitations, and Challengeability, while recognizing contribution rather than demanding impossible proof of attribution. S9 supports early evaluation design, uncertainty, counterfactual reasoning, unintended consequences, and long-horizon evaluation; it does not eliminate political Judgment.

Boundary and uncertainty. Better evidence rules cannot make every consequence observable or neutral. Evidence itself is selected within power relationships. Some material values are partly qualitative, contested, or irreducible to a common measure.

Revision trigger. Narrow the Evidence requirement if valid Accountability can be sustained in defined domains through transparent Judgment and contestation despite persistent empirical incompleteness. Replace the current model if it predictably legitimizes the measurable while excluding material but unmeasured consequences. Reject Verified Value claims where challengeable Evidence cannot exist; do not convert absence of Evidence into proof of absence or success.

Counterclaim 6 - Capability Composition May Destroy More Value Than It Creates

Counterclaim. Every added person, agent, service, model, supplier, or interface creates dependencies and new failure paths. Governance cost, security exposure, opacity, coordination load, and correlated failure can grow faster than nominal execution capacity.

Current response. Hero defines Composition as a governed system relationship, not aggregation. Qualification of units is insufficient; interfaces, observability, revocation, failure propagation, Evidence, and proportional governance matter [S7][S8][S12]. A larger composition is not better merely because it can produce more Output.

Boundary and uncertainty. The Foundation offers no universal way to predict the point at which composition becomes net harmful. Governance can itself become a bottleneck or an illusion of control.

Revision trigger. Narrow the composition proposition if simpler or less governed arrangements consistently outperform governed composition after risk and governance cost are included. Reject the idea that composability is a general advantage if the dominant empirical pattern is increasing unmanageable consequence rather than accountable Value. Hero would then need to treat deliberate non-composition and capability restraint as central rather than exceptional.

Counterclaim 7 - Learning Can Remain Tacit

Counterclaim. Teams, professions, and communities often learn through apprenticeship, practice, story, habit, and shared culture without explicit retained Evidence. Formal memory systems may freeze obsolete Judgment, reduce psychological safety, encourage performative documentation, or lose what experts know tacitly.

Current response. Hero does not equate Organizational Memory with a repository. Chapter 9 requires Evidence, Context, Judgment, and Operational Change because storage without changed behavior is not sufficient. NASA's knowledge policy likewise treats capture as only one part of cultivation, transfer, use, and infusion [S10]. Tacit practice may be one valid carrier of memory.

Boundary and uncertainty. Explicit traceability is not always possible or desirable. Tacit memory may remain reliable in stable, close-knit settings, while formal records can be actively misleading.

Revision trigger. Broaden or replace the four-part memory test if longitudinal Evidence shows that organizations can preserve challengeable learning, transfer it across relevant boundaries, adapt it to changed Context, and retire it when obsolete without retained Evidence or explicit Judgment. Narrow the Organizational Memory claim to high-consequence, high-turnover, distributed, or contested settings if tacit mechanisms are equally reliable elsewhere.

Counterclaim 8 - Existing Models or Informal Practice May Already Be Sufficient

Counterclaim. Product management, agile delivery, systems engineering, professional ethics, safety practice, evaluation, records governance, public administration, and institutional law already address purpose, responsibility, execution, evidence, and learning. Well-led organizations may embody all Hero relationships informally. Hero may be a new vocabulary for established practice rather than a distinct theory.

Current response. Hero makes a narrower claim to distinctiveness: it integrates accountable Value selection, human Primary Accountability, governed Capability Composition, consequence-aware Evidence, and Organizational Memory under conditions of increasingly accessible intelligence-enabled execution. The sources used in this edition show that each adjacent discipline contributes essential concepts; none is evidence that Hero's particular synthesis is necessary.

Boundary and uncertainty. Similarity is not a weakness if the synthesis clarifies real gaps, but renaming alone has no theoretical value. Canon 002 has not yet completed systematic comparison with the strongest existing models.

Revision trigger. Replace or reject Hero as an independent theory if established models, alone or in a coherent existing combination, explain the same failure patterns, preserve the same necessary relationships, and guide Judgment at least as well without Hero-specific concepts. Remove the HeroOS derivation if informal practice sustains these relationships reliably at organizational scale. Retain Hero only as a communication framework if that is the strongest contribution supported by Evidence.

Theory-Level Revision Outcomes

When a material challenge is supported, the Foundation must record one of five outcomes:

- **Retain**: the proposition survives the challenge within its stated scope.
- **Narrow**: the proposition remains credible only in a smaller set of Contexts.
- **Qualify**: the proposition remains, but a condition, exception, or uncertainty must be made explicit.
- **Replace**: another proposition explains the Evidence and counterexamples better.
- **Reject**: the proposition no longer deserves a place in the theory.

Revision must not be hidden as editorial clarification. A future version must identify which proposition changed, what Evidence or reasoning caused the change, and which downstream chapters require reconsideration.

Evaluation Rule

Canon 002 must answer counterclaims with bounded reasoning, comparative Evidence, and visible limitations. Where Evidence is insufficient, the text must state a hypothesis or research question rather than a settled law. A counterexample may expose a scope limit without invalidating the whole theory; repeated accommodation without a possible rejection condition invalidates the claim to falsifiability.

14. Research Agenda and Version History

KEY IDEA

The Foundation should expose what remains unproven so the Canon can become stronger without rewriting its history.

Research Principles

The research agenda exists to expose the Foundation to loss, not merely to collect confirming examples. Studies should preserve adverse cases, distinguish Contexts, compare Hero propositions with credible alternatives, and report uncertainty. Evidence that only shows that a Hero practice can work is insufficient; research must ask whether the proposed relationship is necessary, distinctive, and better than available explanations.

The agenda should combine field cases, longitudinal observation, comparative institutional analysis, controlled or quasi-experimental studies where appropriate, historical countercases, conceptual analysis, and specialist review. No single method can resolve empirical, structural, and normative claims at once.

Research design belongs here at the level of questions and disconfirmation. Formal HeroOS data fields, workflow states, evidence schemas, thresholds, quotas, procedures, and conformance tests remain Canon 006 concerns.

Program 1 – Scarcity Shift Across Domains

Hypothesis. As intelligence-enabled Capability becomes less costly and easier to access in a Context, a greater share of limiting effort moves from producing Output toward selecting Value, accepting Accountability, governing composition, evaluating Consequence, and learning.

Domains. Knowledge work, software, customer operations, science, design, physical operations, healthcare, regulated services, public administration, education, and public-interest work.

Comparison. Compare similar work with materially different access to intelligence-enabled Capability; compare stages within one Value Creation System; compare experienced and novice practitioners; and compare periods before and after a Capability change.

Evidence required. Cost and availability changes, time allocation, throughput and quality, failure and rework, coordination load, authority and escalation patterns, outcome Evidence, stakeholder experience, and the location of unresolved constraints. S1-S4 provide initial bounded examples, not a cross-domain conclusion.

Disconfirming outcomes. The hypothesis is weakened if Capability access rises without changing the distribution of constraint or Judgment, if execution scarcity remains dominant for the complete system, or if governance and verification cost consume the apparent gain.

Required review disciplines. Economics, operations research, labor studies, domain professionals, human-computer interaction, organizational psychology, and affected-worker representation.

Program 2 – Accountability Topologies

Hypothesis. A bounded Value Commitment is more likely to remain answerable when Primary Accountability is identifiable, Authority is aligned or insufficiency is escalated, and independent institutional Accountabilities remain explicit.

Domains. Product development, safety-critical operations, clinical care, public administration, partnerships, boards, courts, scientific collaborations, regulated professions, cooperatives, and crisis response.

Comparison. Compare individual-primary, genuinely shared, institutional, rotating, and formally unowned arrangements. Include cases where an individual center became a blame target or overrode legitimate plural authority.

Evidence required. Decision traceability, challenge and remedy, response to insufficient Authority, stakeholder legitimacy, speed and quality of escalation, consequence allocation, learning after failure, and persistence across personnel change.

Disconfirming outcomes. The one-human-center proposition is weakened if collective or institutional structures preserve final answerability and legitimacy equally or better without accountability diffusion, or if individual-primary structures systematically distort complex causation and independent duties.

Required review disciplines. Law, political theory, corporate governance, public administration, professional ethics, safety governance, organizational sociology, and representatives of affected stakeholders.

Program 3 – Evidence, Delayed Value, and Contested Consequence

Hypothesis. Separating Output, Outcome, Evidence, Value, and Consequence, combined with explicit provenance, relevance, reliability, time boundary, limitations, and challengeability, improves Judgment without pretending to eliminate uncertainty.

Domains. Preventive health, education, climate and infrastructure, public policy, research, brand and trust, risk reduction, social programs, and long-lived commercial products.

Comparison. Compare decisions using immediate proxy metrics, multi-horizon evaluation, qualitative and quantitative evidence, contribution versus attribution reasoning, and stakeholder contestation. Include cases where measurement caused goal displacement or excluded material interests.

Evidence required. Baselines, causal or contribution arguments, unintended effects, distribution of benefits and harms, revisions over time, challenges to the evidence, and decisions changed by later information. S9 provides evaluation principles but does not validate Hero's evidence model.

Disconfirming outcomes. The model is weakened if its distinctions do not improve decisions, if they create unjustified confidence, or if transparent professional or democratic Judgment works better in domains where Evidence remains structurally incomplete.

Required review disciplines. Evaluation science, statistics, causal inference, epistemology, audit, behavioral science, public policy, qualitative research, and stakeholder or community review.

Program 4 - Capability Across Human and Non-Human Executors

Hypothesis. Capability can be defined technology-agnostically through bounded conditions, repeatability, verification, qualification, observability, and fitness for Purpose, while preserving important differences among people, organizations, machines, services, and mixed units.

Domains. Expert professional work, skilled trades, AI-assisted knowledge work, autonomous systems, shared services, external suppliers, teams, and hybrid human-machine operations.

Comparison. Compare equivalent functions performed by different executor types and compositions. Examine creative, relational, non-repeatable, and tacit work that may resist stable Capability descriptions.

Evidence required. Performance distributions, boundary conditions, Context dependence, degradation, recoverability, handoff quality, qualification validity, human effects, and failure modes. Reliability must not be treated as the sole measure of fitness.

Disconfirming outcomes. The common concept is weakened if it erases morally or operationally decisive differences, cannot represent non-repeatable expertise, or produces qualifications that fail under ordinary Context change.

Required review disciplines. Systems engineering, human factors, AI assurance, professional education, labor studies, cognitive science, safety, accessibility, and domain specialists.

Program 5 - Capability Composition, Failure Propagation, and Governance Cost

Hypothesis. Composed Capability creates legitimate scale only when relationship value exceeds interface, coordination, assurance, and failure-propagation cost, and when the composition remains observable and revocable in proportion to Consequence.

Domains. Software and AI systems, supply chains, healthcare pathways, public-service ecosystems, financial operations, emergency response, and cross-organizational programs.

Comparison. Compare small and large compositions, centralized and distributed arrangements, tightly and loosely coupled systems, and governed versus nominal aggregation. Include successful non-composition and deliberate simplification.

Evidence required. Dependency maps, interface failures, correlated incidents, governance effort, security and safety exposure, recovery time, Value produced, downstream Consequence, and cases where qualified units produced an unqualified system. S7, S8, and S12 supply adjacent systems reasoning, not proof of the Hero proposition.

Disconfirming outcomes. The proposition is weakened if the proposed governance relationships do not predict or reduce material failures, or if governance cost routinely exceeds the Value preserved. A consistent advantage for simpler arrangements would require composition to become a narrower exception.

Required review disciplines. Systems and safety engineering, cybersecurity, resilience, operations research, complexity science, architecture, supplier governance, and frontline operators.

Program 6 - Learning and Organizational Memory

Hypothesis. Learning becomes Organizational Memory when Evidence, Context, Judgment, and Operational Change remain available enough to influence future action, while obsolete memory can lose active authority through review, supersession, retirement, archival, or disposal.

Domains. High-turnover teams, professional practice, incident learning, long-lived programs, public institutions, regulated operations, research organizations, and communities of practice.

Comparison. Compare repositories, apprenticeship, narrative practice, standards, training, embedded controls, active-memory limits, and no-formal-memory settings. Include over-retention, forgotten lessons, stale rules, and undocumented but resilient tacit practice.

Evidence required. Reuse in later decisions, Context-fit checks, behavioral change, transfer across people and boundaries, challenge and correction, retirement of obsolete Judgment, search and maintenance burden, and recurrence of preventable failure. S10 and S11 support knowledge-use and exit concerns but do not prove the four-part test.

Disconfirming outcomes. The test is weakened if tacit learning preserves and updates organizational Judgment equally well in relevant Contexts, if explicit Evidence does not improve future action, or if Active Memory controls create more loss than clarity.

Required review disciplines. Knowledge management, records governance, organizational learning, anthropology, history, information science, professional education, privacy, and legal review.

Program 7 - Distinctiveness and Existing-Model Sufficiency

Hypothesis. Hero's integration of Purpose, Responsibility, Capability, Value, Learning, Accountability, Context, Value Commitment, Evidence, Consequence, and Organizational Memory explains a meaningful class of failures not handled as coherently by existing models or informal practice.

Domains and comparators. Product management, agile and lean practice, systems engineering, theory of constraints, socio-technical systems, professional ethics, safety management, evaluation, organizational learning, public-value theory, governance, institutional design, and ordinary high-performing teams.

Evidence required. Concept mapping, strongest-form comparison rather than comparison with weak practice, paired cases, explanatory coverage, predictive usefulness, decision quality, adoption cost, vocabulary burden, and cases where Hero adds no value.

Disconfirming outcomes. Hero should be replaced as an independent theory if an existing model or established combination provides equal or better explanation and guidance with fewer assumptions. HeroOS should lose its claimed necessity if informal practice preserves the complete relationships reliably at scale. A finding that HERO is useful only as a mnemonic should narrow the Canon accordingly.

Required review disciplines. Organization theory, product and agile scholarship, systems thinking, management history, philosophy of science, institutional economics, public administration, and experienced practitioners from the compared traditions.

Whole-Foundation Evidence Expansion Agenda

Before publication review, the Foundation requires a broader evidence base than S1-S12. Expansion should prioritize:

- systematic and peer-reviewed evidence on AI, automation, productivity, work design, and task redistribution;
- accountability theory across law, democratic governance, professions, corporations, safety, and collective action;
- stakeholder legitimacy, public Value, economic sustainability, and distributional consequence;
- capability, competence, expertise, delegation, automation, and human-machine teaming;
- systems composition, complexity, resilience, security, safety, and governance cost;
- evaluation, causal inference, measurement politics, delayed outcomes, and qualitative Evidence;
- organizational learning, tacit knowledge, records, institutional memory, forgetting, and knowledge retirement;
- strongest-form comparisons with product, agile, lean, systems, institutional, and professional models;
- adverse and failed cases, especially where Hero-like language masked weak Authority, blame transfer, surveillance, bureaucracy, or unverified Value;
- non-English, non-Western, public-interest, small-organization, and Global South Contexts to test whether the theory embeds narrow institutional assumptions.

Evidence expansion must not become citation accumulation. Each source must identify which proposition it supports or challenges, its Context and limitations, and whether it changes the theory-level confidence.

Specialist Review Agenda

A complete Foundation is not the same as a final multidisciplinary theory. Ongoing review should include, at minimum, expertise in ethics, law, political and institutional theory, economics, organization science, product and agile practice, systems and safety engineering, AI governance and human factors, evaluation and causal inference, knowledge and records governance, labor and professional practice, public administration, and bilingual Chinese-English editorial review.

Affected stakeholders are not interchangeable with technical reviewers. Where a proposition allocates consequence or defines Value for a group, representatives of that group must be able to challenge the framing and Evidence.

Versioning Principles

Version history exists to preserve the development of the theory, including correction and rejection.

- English remains the Source of Truth; Chinese must preserve equivalent claim strength and must not silently resolve ambiguity.
- A version must distinguish editorial changes from substantive changes to propositions, scope, definitions, relationships, counterclaims, or revision rules.

- Substantive revision must identify the triggering Evidence or argument and the downstream chapters reconsidered.
- Earlier versions must remain historically recoverable; a new version must not rewrite an earlier claim as though it had never existed.
- Draft completion, theory acceptance, publication authorization, and HeroOS conformance are different decisions.
- Canon 002 versioning records theory development. It does not define Canon 006 implementation workflows, approval roles, record fields, lifecycle states, or release controls.
- No version label can substitute for explicit publication authority.

Version History

Version	Date	Status	Notes
0.1	2026-07-27	Draft for Review	Initial structure and proof-burden draft; not approved for publication
0.2	2026-07-27	Draft for Review	Continuous full-theory draft completed for Chapters 1-14 with bounded counterclaims, falsification rules, and research program; adversarial, specialist, bilingual, evidence, and publication review remain required
1.0	2026-07-27	Founding Edition	First approved and formally published Foundation baseline; future evidence may retain, narrow, qualify, replace, or reject its propositions through visible version history

Appendix A - Research Sources

KEY IDEA

Sources discipline Hero's Judgment; they do not replace it.

The sources below support, limit, or challenge propositions in this Foundation. Inclusion does not mean that Hero adopts every conclusion of a source. Each source must be read within its stated Context and limitation.

S1 - Stanford Institute for Human-Centered AI, AI Index Report 2025

Evidence used: inference cost for a GPT-3.5-equivalent MMLU performance level fell from USD 20 per million tokens in November 2022 to USD 0.07 by October 2024. This supports declining access cost for a defined class of intelligence-enabled execution.

Limitation: benchmark-equivalent inference cost is not successful organizational execution, realized Value, or reduced physical scarcity.

Source: Stanford AI Index 2025

S2 - Erik Brynjolfsson, Danielle Li, and Lindsey R. Raymond, Generative AI at Work

Evidence used: the study of 5,179 customer-support agents found a 14% average increase in issues resolved per hour and a 34% increase for novice and lower-skilled workers, with minimal effect on experienced and highly skilled workers.

Limitation: one operational setting does not establish a universal productivity effect or prove that Execution is no longer a bottleneck.

Source: NBER Working Paper 31161

S3 - METR, Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity

Evidence used: the randomized study reported that experienced developers working in familiar repositories took 19% longer when AI tools were allowed. This supports the counterclaim that additional Capability can increase execution time in high-Context work.

Limitation: METR limits generalization and later stated that the early-2025 result no longer represents current tools.

Source: METR Early-2025 Developer Study

S4 - METR, We Are Changing Our Developer Productivity Experiment Design

Evidence used: the update reports participant and task selection effects, concurrent-agent measurement difficulty, and wider AI adoption. It supports Context, time boundaries, and measurement humility.

Limitation: the update does not provide a reliable single productivity estimate.

Source: METR Uplift Update

S5 - National Institute of Standards and Technology, AI Risk Management Framework 1.0

Evidence used: the framework treats governance as cross-cutting and emphasizes human centricity, social responsibility, sustainability, Context, intended and unexpected impacts, and responsibilities across AI actors.

Limitation: it is a voluntary risk-management framework, not proof that Hero's specific Accountability theory is correct.

Source: NIST AI RMF 1.0

S6 - Business Roundtable, Statement on the Purpose of a Corporation

Evidence used: the statement addresses customer Value, employees, suppliers, communities, environment, and long-term shareholder Value. It supports plural stakeholder Value within a commercial system.

Limitation: it is a voluntary corporate statement and does not resolve stakeholder conflict or establish Hero's Value theory.

Source: Business Roundtable Statement

S7 - National Institute of Standards and Technology, Engineering Trustworthy Secure Systems

Evidence used: NIST SP 800-160 Volume 1 Revision 1 grounds assurance in objective Evidence and treats complexity, dependencies, uncertainty, verification, lifecycle, and compositional trustworthiness as system concerns.

Limitation: systems-security engineering guidance does not establish Hero's Value theory or Primary and Independent Accountability allocation.

Source: NIST SP 800-160 Volume 1 Revision 1

S8 - NASA System Safety Handbook, Volume 1

Evidence used: the handbook treats systems holistically and addresses interactions, dependencies, combinatorial effects, complexity, scenario analysis, and uncertainty. It supports system-level failure-propagation reasoning.

Limitation: NASA safety methods and risk tolerance cannot be transferred mechanically to every Hero Context.

Source: NASA System Safety Handbook

S9 - HM Treasury and UK Government Analysis Function, Magenta Book

Evidence used: the guidance calls for evaluation before, during, and after implementation, including Theory of Change, Baseline, uncertainty, failure, unintended consequence, counterfactual, contribution or attribution, and long-horizon reasoning.

Limitation: UK public-policy guidance does not define Hero Accountability and must be adapted proportionately in other Contexts.

Source: The Magenta Book

S10 - NASA Knowledge Policy for Programs and Projects

Evidence used: the policy distinguishes cultivating, identifying, capturing, retaining, using, and sharing knowledge, and calls for lessons to be infused into standards, training, methodologies, and operations.

Limitation: NASA policy does not prove Hero's four-part Organizational Memory validity test or a universal memory lifecycle.

Source: NASA NPD 7120.6A

S11 - US National Archives and Records Administration, Guide to Records Inventory, Scheduling, and Disposition

Evidence used: the guidance distinguishes temporary and permanent records, retention and destruction, superseded authority, and schedule review. It supports explicit memory exit and review distinctions.

Limitation: records management is not identical to Organizational Memory; changing decision influence and behavior requires more than record disposal.

Source: NARA Records Scheduling Guide

S12 - NASA Systems Engineering Handbook

Evidence used: the handbook defines systems through elements and relationships, describes recursive and iterative processes, distinguishes Verification from Validation, and notes that necessary functions may be performed by different roles with proportional formality.

Limitation: technical systems-engineering guidance does not establish Hero's Value, Accountability, team topology, or organizational theory and must not become a HeroOS process specification.

Source: NASA Systems Engineering Handbook